

AGHAMazing Quest: CMS

Project Documentation Submitted to the Faculty of the
School of Computing and Information Technologies

Asia Pacific College

For Fulfillment of the Requirements for
Introduction to Systems and Design for IT
MSYADD1

By

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MSYADD1 Final Paper contents:

- **Charter**

Name

AGHAMazing Quest (Content Management System)

Purpose

The AGHAMazing Quest Content Management System (CMS) aims to enhance the delivery, control, and flexibility of content management within the AGHAMazing Quest mobile application. Currently, the challenge lies in the inflexibility of the existing CMS, which forces DOST-STII to depend on developers for even minor updates. This project will provide DOST-STII with a more adaptable and user-friendly CMS, enabling staff to independently manage and update their content while maintaining ownership of both the CMS and their data.

High-level project description

The AGHAMazing Quest CMS application is for web users. It's designed to improve content management within DOST-STII. The app will have important features like:

- Content Management: Create, edit, publish, delete content; support for rich text, images, files
- Search & Indexing: Fast, accurate search across all content
- User Management: Role-based access control (admin, editor, viewer, etc.)
- Workflow: Draft, review, approve, publish, archive
- Versioning: Track changes, revert to previous versions
- Media Management: Upload and manage images, videos, documents
- API Integration: RESTful or GraphQL APIs for app integration
- Analytics: Basic reporting on content usage and activity

The project will go through different phases: planning, designing the interface, building the web app, testing it with real users, and finally presenting it during our final defense.

High-level milestone schedule

Milestone	Estimated Week
Development Completion	Week 7
Unit Testing Completion	Week 8
Integration Testing Completion	Week 9
System Testing Completion	Week 10
UAT Completion	Week 11
Code Freeze Period	Week 12
Deployment Date	Week 13
Release Sign-off	Week 14

Rough cost estimate and budget

The expected cost for this project would be around P60,000

- Data or internet use during development
- Printed materials / documents for training materials
- Tools or software / Subscriptions

Most of the necessary tools and software would be free and open source as this is what DOST-STII requests. Other expenses would be shouldered by the developers such as travel from APC to their site and food costs.

Stakeholders

- DOST-STII Staff (users of the system)
- Other government sectors (possible partners in the future)
- Future researchers

Project manager

Joshua Sta.Ana, Group Leader

Project manager's responsibilities

- Make sure the group is on track with the timeline
- Assign tasks to each member and follow up on progress

- Communicate with the instructor when needed
- Assist in development, testing, and documentation
- Organize the presentation and final defense

Project manager's authority

- Decide on which tasks should be done first
- Reassign tasks if someone can't finish on time
- Handle the group's budget and expenses
- Solve problems or misunderstandings in the group
- Make final decisions with team input

Formal declaration of the sponsor's support

Sponsor Name: Jose Eugenio L. Quesada

Position: Project-Based Learning Instructor

Support Statement:

As the instructor for the Project-Based Learning course, I support the AGHAMazing quest CMS project made by the student group. This project has the potential to improve the content management experience at DOST-STII and shows the students' ability to solve real-world problems through technology. I encourage the group to complete the project with teamwork, effort, and creativity.

- Objectives

AGHAMazing Quest CMS Case Study Objectives

The following represents notes taken from an interview about project objectives with representatives from DOST-STII, the project sponsor:

The current content management process for the AGHAMazing Quest mobile application is limited by lengthy update procedures and an inflexible AR tour app. This results in delayed content updates, limited flexibility, and increased operational costs for DOST. The primary objective is to modernize content management by developing a web-based Content Management System (CMS). By achieving this objective, the project aims to significantly reduce content update time and DOST will have the ability to upload the content on their own.

The CMS will provide DOST-STII staff with the ability to independently create, edit, publish, and manage content for the mobile application. Features such as content management (create, edit, publish, delete content with support for rich text, images, files), search & indexing capabilities, user management with role-based access control, content workflow management (draft, review, approve, publish, archive), versioning, media management, API integration with the mobile application, and basic analytics will improve content delivery, control, and flexibility.

An efficient CMS will support DOST-STII's mission by ensuring timely and relevant content is available for visitors using the AGHAMazing Quest mobile application, ultimately enhancing the overall visitor experience.

The project will follow a structured development approach with milestones from Week 7 to Week 14, as outlined in the project charter, including development completion, various testing phases, code freeze, deployment, and final release sign-off.

Categorized Objectives

Objective	Category	Measure
Modernize content management with a web-based CMS for the AGHAMazing Quest application	Performance	Successful CMS launch and staff adoption
Reduce content update time significantly compared to current processes	Performance	Percentage reduction in time required for content updates
Independent Upload of content	Resources	Percentage reduction in third-party developer involvement
Implement core CMS features including content management, search & indexing, user management, workflow,	Performance	Number of features successfully implemented

versioning, media management, API integration, and analytics		
Ensure seamless integration between CMS and the AGHAMazing Quest mobile application	Customer Focus	Successful API integration and user satisfaction scores
Complete MVP by December 2025 and full release by June 2026	Resources	Delivery within timeline

- **Scope**

Project goal and objectives

What is the project supposed to achieve.

1. Enable DOST-STII staff to independently manage and update content through an easy-to-use CMS within 4 months.
2. Reduce the content update process duration by 60% within 4 months.
3. Provide analytics on user engagement within 3 months of implementation to guide content improvements.

Project Boundaries

Within scope: What is included in the project.

- Content Management: Create, edit, publish, delete content; support for rich text, images, files
- Search & Indexing: Fast, accurate search across all content
- User Management: Role-based access control (admin, editor, viewer, etc.)
- Workflow: Draft, review, approve, publish, archive
- Versioning: Track changes, revert to previous versions
- Media Management: Upload and manage images, videos, documents
- API Integration: RESTful or GraphQL APIs for app integration
- Analytics: Basic reporting on content usage and activity

Out of scope: What is excluded from the project.

- Development or redesign of the AGHAMazing Quest AR mobile app itself.
- Public access to the CMS (restricted to authorized DOST staff).
- Dedicated mobile app version of the CMS.

- AR hardware or marker setup beyond integration points.
- Creation of content to be uploaded to the CMS

Project Deliverables

The tangible items produced by the project.

- Fully functional, responsive web-based CMS.
- RESTful APIs for content synchronization with the mobile app.
- Role-based user management system.
- Content analytics and reporting tools.
- Media library with search, tagging, and bulk upload features.
- Documentation: user manuals, technical documentation, and training materials.

Success Criteria

How you determine that the project and deliverables have been completed successfully.

- DOST-STII staff can independently update content without developer assistance.
- Content updates made by staff are reflected in the AGHAMazing Quest mobile app within minutes of publishing.

- Analytics dashboards accurately report engagement metrics.
- System passes all functional, usability, and security testing.
- Project completed within approved timeline and budget.

Project Assumptions

Information that was not known at the time.

- DOST-STII staff have internet access and compatible devices to use the CMS.
- The mobile app development team will support API integration.
- DOST-STII will provide branding guidelines and content for initial uploads.
- Adequate training will be provided to CMS users before launch.

Project Constraints

Limitations such as time, budget, etc.

- Limited to web-based access; no dedicated CMS mobile application.
- Integration schedule dependent on the mobile app development team's timeline.
- Budget constraints as a government-funded project.
- Project timeline must align with DOST's internal approval and review processes.

- Stakeholder analysis

Name	Department / Company	Position	Advisers	Objectives, Requirements, Interests	Influence	Project Contribution	Resistance
DOST-STII Staff		Content Managers		Update AR app content efficiently/ Weekly updates via email; UAT sessions	High		
Joshua Sta.Ana	Student/APC	Project Lead/Developer	Jojo F. Castillo	Deliver CMS on time and within scope/Daily standups; sprint reviews	High		
Ervyne Anthony V. Chan	Student/APC	QA Engineer		Ensure system reliability and performance/Defect reports; test case reviews	Medium		
Francis Frederic Encomienda	Student/APC	CMS Developer	Ryan John Perez/	Backend logic and API integration/Technical sync meetings	High		
Marianne Ellise P. Lama	Student/APC	Documentation Lead	James Raphael L. Ibay	Clear user manuals and guides/Biweekly documentation reviews	Medium		
Frances Sophia Lepso	Student/APC	DevOps Engineer		Objectives: Ensure reliable infrastructure, CI/CD pipelines, and deployment processes	High		
				Requirements: Weekly technical syncs, documentation reviews, and incident response coordination.			
				Interest: Focuses on system reliability, deployment automation, and cloud infrastructure (AWS/Docker).			
Ms. Irene Brillo		DOST-STII Unit Head		Compliance with organizational goals/Monthly progress presentations	High		
Mobile App Developers	Student/APC	External Team		Seamless API integration with AR app/API documentation; integration workshops	Medium		
Visitors/Users		End-users of AR app		Engaging, up-to-date content/Feedback surveys; analytics reports	Low		

OpenProject / Jira Activities

OpenProject Budget

1.0 Initialization Phase

Added by Ervyn Anthony Chan 2 months ago.

Cost type

Budget

Fixed date

08/28/2025

Spent (ratio)

0% Total progress

Description

UNITS

Planned unit costs				Actual unit costs			
UNITS	COST TYPE	COMMENT	BUDGET	WORK PACKAGE	UNITS	COST TYPE	COSTS
1.00	Internet Monthly Subscription Php1K	Developer's Internet	PHP 1,000.00				PHP 0.00
1.00	Utilities monthly - Electricity	Developer's Electricity	PHP 100.00				
3.00	User Training expenses daily Php 1K		PHP 3,000.00				
1.00	Computer Equipment	Computer / Laptop	PHP 50,000.00				
1.00	Computer Equipment	Computer / Laptop	PHP 50,000.00				
1.00	Computer Equipment	Computer / Laptop	PHP 50,000.00				
1.00	Computer Equipment	Computer / Laptop	PHP 50,000.00				
1.00	Computer Equipment	Computer / Laptop	PHP 50,000.00				
			PHP 254,100.00				

LABOR

Planned labor costs				Actual labor costs			
HOURS	USER	COMMENT	BUDGET	WORK PACKAGE	HOURS	USER	COSTS
8.00 hours	Ervyn Anthony Chan	2 Hours / Week	PHP 800.00				PHP 0.00
8.00 hours	Francis Encomienda	2 Hours / Week	PHP 800.00				
8.00 hours	Marianne Ellise Lama	2 Hours / Week	PHP 800.00				
8.00 hours	Frances Lepso	2 Hours / Week	PHP 800.00				
8.00 hours	Joshua Sta Ana	2 Hours / Week	PHP 800.00				
			PHP 4,000.00				

Jira Work Packages

Data Flow Diagram – Joshua Sta Ana

Entity Relationship Diagram – Joshua Sta Ana + Group

Use Case Diagram – Eryne Anthony Chan

Test Cases – Frances Sophia Lepso + Francis Frederick L. Encomienda

Requirement Traceability Matrix – Group

SWOT Analysis – Marianne Ellise Lama

Design Thinking Board – Marianne Ellise Lama

Jira Backlogs

☐ **Sprint 01: Documentation** 18 Sep – 2 Oct (8 work items)

Fix the missing and unedited documents

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Complete sprint

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☒ AQ-3 Data Flow Diagram	DOCUMENTATION	IN PROGRESS	0	JA
☒ AQ-2 Use Case Fully Dressed	DOCUMENTATION	DONE	-	AC
☒ AQ-4 Entity Relationship Diagram	DOCUMENTATION	DONE	-	JA
☒ AQ-1 Use Case Diagram	DOCUMENTATION	DONE	-	AC
☒ AQ-5 Test Cases	DOCUMENTATION	DONE	-	JA
☒ AQ-6 Requirement Traceability Matrix	DOCUMENTATION	DONE	-	JA
☒ AQ-7 SWOT Analysis	DOCUMENTATION	DONE	-	EB
☒ AQ-8 Design Thinking Board	DOCUMENTATION	DONE	-	EB

+ Create

☐ **Sprint 02: Urgent Features** 7 Oct – 9 Oct (4 work items)

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Complete sprint

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☒ AQ-10 [System Admin] [Manage Users] - As a system administrator, I need to perform CRUD operations on user accounts	SYSTEM FEATURES	IN PROGRESS	-	
☒ AQ-13 [System Admin] [RBAC] - As a system administrator, I need to be able to differentiate permissions for each role to maintain security.	SYSTEM FEATURES	IN PROGRESS	-	
☒ AQ-11 [System Admin] [Manage Content] - As a system administrator, I need to perform CRUD operations on content being uploaded	SYSTEM FEATURES	TO DO	-	
☒ AQ-12 [System Admin] [API Management] - As a system administrator, I need to be able to manage APIs.	SYSTEM FEATURES	TO DO	-	

+ Create

☐ **Backlog** (6 work items)

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Create sprint

...

☒ AQ-14 [System Admin] [Versioning] - As a system administrator, I need to be able to create versions to track and revert back to previous versions when needed.	SYSTEM FEATURES	TO DO	-	
☒ AQ-15 [Content Creator] [Uploading] - As a content creator, I want to be able to upload my created content	SYSTEM FEATURES	TO DO	-	
☒ AQ-16 [Content Editor] [Editing] - As a content editor, I want to have the option to edit the uploaded content.	SYSTEM FEATURES	TO DO	-	
☒ AQ-17 [Content Approver] [Approval] - As a content approver, I want to have the option to accept or reject uploaded content.	SYSTEM FEATURES	TO DO	-	
☒ AQ-18 [Content Approver] [Publishing] - As a content approver, I want to have the option to publish approved content.	SYSTEM FEATURES	TO DO	-	
☒ AQ-19 [Users] [Analytics] - As a DOST-STII staff, I want to see the different metrics of the AGHAMazing Quest Application.	SYSTEM FEATURES	TO DO	-	

+ Create

Jira Board AY-2025-2026 T1 Sprint 01

Spaces

AGHAMazing Quest

SummaryTimelineBacklogBoardGoalsAll workDevelopmentCode

Q Search board

Filter

TO DO 2

[System Admin] [Manage Content] - As a system administrator, I need to perform CRUD operations on content being uploaded

SYSTEM FEATURES

AQ-11

[System Admin] [API Management] - As a system administrator, I need to be able to manage APIs.

SYSTEM FEATURES

AQ-12

+ Create

IN PROGRESS 3

Data Flow Diagram

DOCUMENTATION

AQ-3

[System Admin] [Manage Users] - As a system administrator, I need to perform CRUD operations on user accounts

SYSTEM FEATURES

AQ-10

[System Admin] [RBAC] - As a system administrator, I need to be able to differentiate permissions for each role to maintain security.

SYSTEM FEATURES

AQ-13

DONE 7

Use Case Fully Dressed

DOCUMENTATION

AQ-2



Entity Relationship Diagram

DOCUMENTATION

AQ-4

Use Case Diagram

DOCUMENTATION

Documentation

AQ-1

Test Cases

DOCUMENTATION

AQ-5

Requirement Traceability Matrix

DOCUMENTATION

AQ-6

SWOT Analysis

DOCUMENTATION

AQ-7

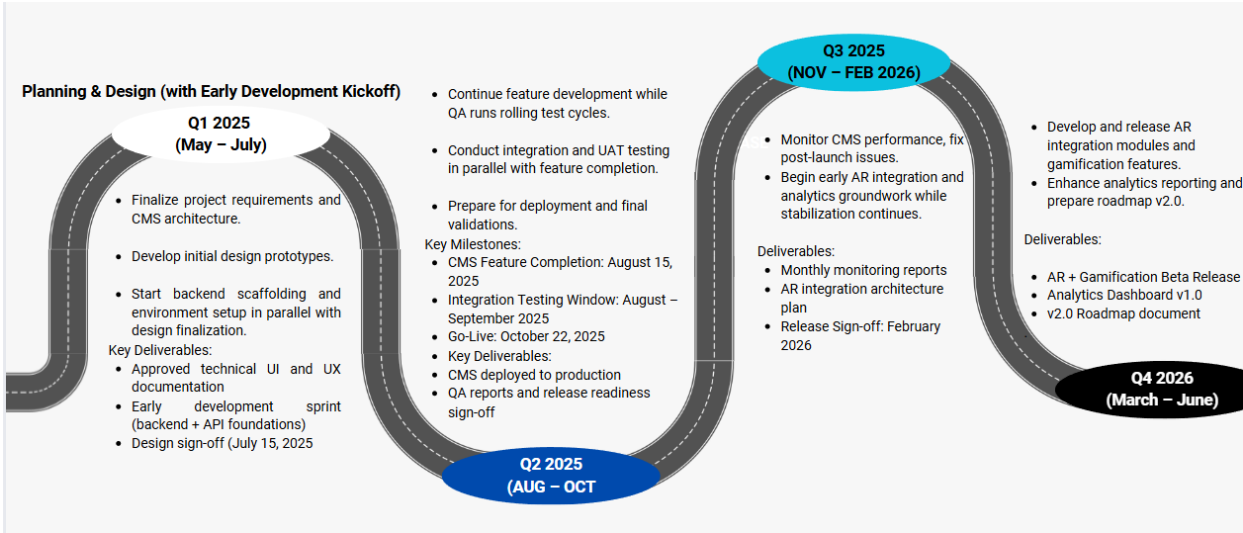
Design Thinking Board

DOCUMENTATION

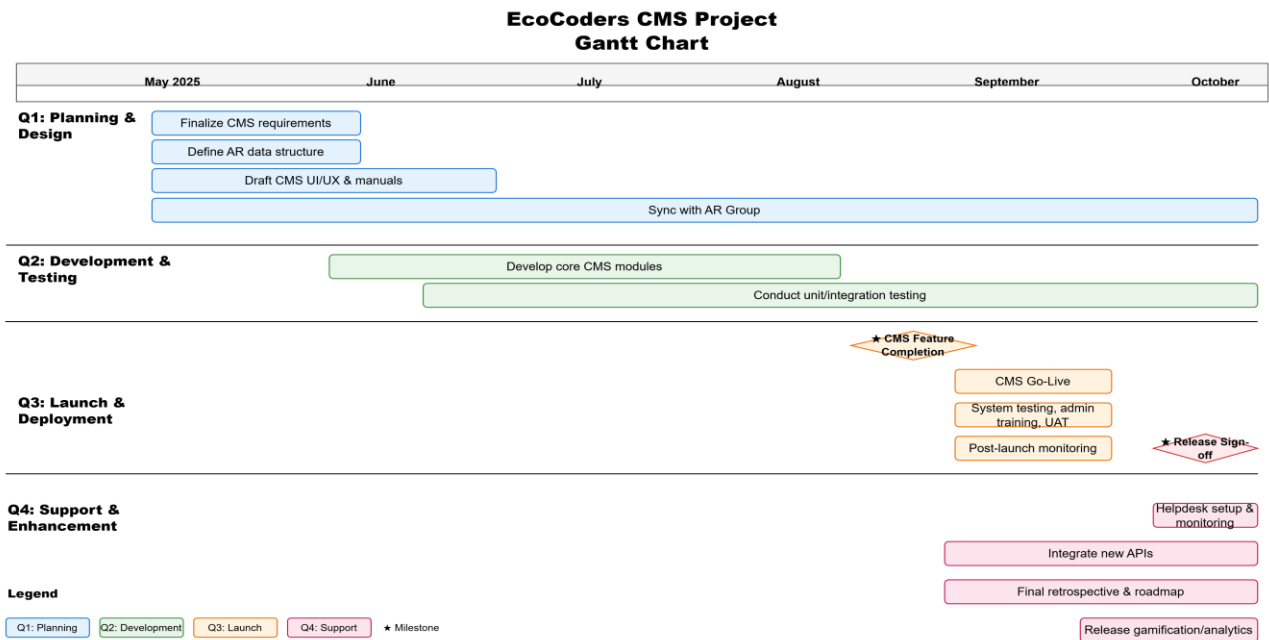
AQ-8

Openproject Activities

- Roadmap



- Gantt chart



Design Thinking output (5 Stages):

- Stage 1 - Empathize

Introduction

The first step of the design thinking process is to identify and address the problem at hand. We put ourselves in the client's perspective on what they do and what they need to accomplish their goals.

In DOST-STII, we interviewed three people, Ms Irene, Sir Jonathan Abalon and Ms Grace Ann Valondo. We also saw how the staff interact with their current software.

The DOST-STII building was filled with numerous AR Markers which the application used to display the necessary details and information, and all these details and information are fixed and cannot be edited with ease. This was because the previous developer of the current DOST-STII AGHAMazing Quest application did not provide the source code to the application, which makes it not editable.

- The following are the process we did to accomplish this stage:
- Asking the 5 whys – Completed for all stakeholders
- Asking open-ended questions – Completed for all stakeholders
- Creating Personas – Completed
- Creating an Empathy Map – Completed
- Pain-Gain Analysis – Completed

The initial visit to the DOST-STII building was with four (4) group members, so we were lacking one (1) group member on the initial visit. This visit consisted of a tour of most of the DOST-STII building, how the AR Markers work, how the application worked and what the issues were within their current application.

During this visit, we were able to determine what was their main concern on why they wanted to change or edit their current application, what features they wanted to add or improve, how they wanted to implement it and where the new application would be implemented or created. This provided the necessary feedback to help us determine what is needed.

- **Interview with the Client:**



- **Flow of the interview with Ms. Irene:**

Question	Answer
How does the AHAMazing Quest Tour App project align with DOST's mission and priorities?	<i>"It was during the pandemic; through the AGHAMazing Quest tour app, we've realized that the project will place a long-term hold and key in our agencies' digital transformation using modern software applications to promoting science and technology to the public as our mission starting from DOST-STII."</i> <i>"However, we are still fully dependent on the developers for its update and maintenance back then that costs us more budget & delay time."</i> <i>" The development of the web app CMS will become a game changer towards our priorities by effectively managing the content of the mobile app as our own, and we can focus on what matters first by possibly allocating more budget to other projects."</i>
What is the primary goal of the AGHAMazing Quest CMS?	<i>"To eliminate the long-term dependency for the developer support on the update of the mobile app contents"</i>
"What style or tone should the web app convey?	<i>"Modern, innovative and functional"</i>
What is the desired domain name or branding approach?	<i>"Can be AGHAMazing Quest CMS" and it should be aligned with DOST branding guidelines"</i>
Who are the main users of the CMS web app (students, tourists, educators, Staff, etc.)?	<i>"Our DOST admins and staffs"</i>
Are there specific different user roles (admins, content moderator)?	<i>"Ideally, we want our users in the system to have a role of Encoder, Reviewer & Approver" however, we are very open for your team's innovative approach in developing the web app, other roles, if relevant and functional, will be reviewed and appreciated."</i>
Why does updating content from the AGHAMazing Quest Tour App take longer?	<i>We need close coordination with the third-party developer team, which sets deadline and approval</i>
How does it make you feel?	<i>"There were odd times, more often we feel left behind and have no total ownership of the AGHAMazing tour AR app project."</i>

- It was an afternoon, during our visit to the DOST complex alongside with other PBL teams, we met with our SPOC's and we held a walk-and-talk meeting at the DOST-STII building. After a while, we sat in their library where the above questions were asked.
- Our SPOCs were 3 of the few DOST-STII staff, who are present, we then discussed further details on how they want a working CMS for their application to be created and developed.

- While with Ms Irene, Sir Jonathan an alumnus of APC took time asking questions from us
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8/31/2022

- **Flow of the interview with Sir Jonathan:**

Question	Answer
How does the AGHAMazing Quest CMS align with DOST-STII's digital transformation goals?	<i>"The CMS is crucial for our digital transformation as it gives us control over our content. Without it, we're stuck in a cycle of dependency that slows down our ability to respond to public needs and showcase new scientific developments."</i>
What is the primary goal of the AGHAMazing Quest CMS from your perspective?	<i>"From my perspective as a technical staff member, the CMS should allow us to update content without waiting weeks for developer availability. We need immediate control over our information flow."</i>
Why does the current update process take so long?	<i>"The current process requires us to send requests to the developers, wait for them to schedule it, then wait for implementation and testing. A single small change can take 2-3 weeks."</i>
What specific features do you consider essential for the CMS?	<i>"We need a simple interface where we can add text, images, and videos to our AR markers. Version control is important so we can revert if needed, and the workflow system with Encoder-Reviewer-Approver roles is critical for our content approval process."</i>
How do you envision the CMS improving your daily work?	<i>"It would save us tremendous time and give us flexibility to respond quickly to events. For example, if there's a new scientific discovery we want to highlight, we could update the content immediately rather than waiting."</i>
What technical constraints should we be aware of?	<i>"The CMS needs to integrate with the existing mobile app through APIs. We're limited by the AR marker system already in place, so the CMS must work with that infrastructure."</i>
How do you measure success for this project?	<i>"Success would be measured by how quickly we can update content - ideally within minutes rather than weeks - and by how many staff members feel comfortable using the system without too much of developer assistance."</i>

Flow of the interview with Sir Allan:

Question	Answer
How do you currently manage content updates for the AGHAMazing Quest app?	<i>"We have to compile all the changes we want in a document, send it to the developers, and wait for them to implement it. Sometimes we have to follow up multiple times to get simple changes done."</i>
What's the most frustrating part of the current content management process?	<i>"The most frustrating part is when we need to make urgent updates, like correcting information or adding timely content for an event, and we have to wait because the developers are busy with other projects."</i>
How would your work change if you had direct control over content updates?	<i>"I could immediately update exhibits when new information comes in, fix errors right away, and keep our content fresh and relevant. This would significantly improve our visitor experience."</i>
What specific types of content do you need to update most frequently?	<i>"We frequently update exhibit descriptions, add new images, and sometimes add short videos. Occasionally we need to add completely new exhibits when new technologies are installed."</i>
How important is the approval workflow (Encoder-Reviewer-Approver) to your process?	<i>"It's extremely important. We need to maintain quality control and ensure that all content meets our standards before it goes live. The three-tier system could reduce mistakes before it goes public."</i>
What do you think would make the CMS intuitive for your other non-technical staff or colleague	<i>"Simple drag-and-drop functionality, clear instructions, and a preview feature that shows exactly how the content will appear in the mobile app would make it much easier for us to use."</i>
How do you see the CMS supporting DOST-STII's mission to promote science and technology?	<i>"By allowing us to keep content current and engaging, we can better fulfill our mission of making science accessible to the public. Timely updates mean visitors get the most relevant information during their tour."</i>

Open-Ended Questions Responses

For Ms. Irene Brillo:

"Can you describe a specific situation where the lack of content control affected your team's ability to fulfill DOST-STII's mission?"

"During the National Science and Technology Week last year, we discovered an error in one of our exhibit descriptions about renewable energy technologies. Because we couldn't update the content ourselves, we had to wait two weeks for the developers to make this correction.

This meant visitors received incorrect information during a critical outreach period, which directly contradicted our mission of accurately promoting science and technology to the public. The delay prevented us from maintaining the credibility and accuracy that DOST-STII must uphold as a government science institution."

"How would having a CMS change the way your department plans and executes its public engagement activities?"

"With a CMS, we could become much more responsive and dynamic in our planning. Currently, we must schedule content updates months in advance because of developer dependency. With direct control, we could create timely content around current events - like when there's a major scientific breakthrough or during special science observances.

We could also adjust content based on visitor feedback collected during tours, creating a more responsive and engaging experience. Most importantly, we could develop content calendars that align with our strategic priorities rather than being constrained by developer availability."

"What would 'success' look like for you when this CMS is implemented?"

"Success would mean that within three months of implementation, all DOST-STII staff with appropriate roles can independently update content without developer assistance. Specifically, I'd consider it successful if content updates that previously took 2-3 weeks can be completed within 24 hours. Another measure of success would be when our staff feels complete ownership of the content - when they can make updates during their regular workday rather than having to plan weeks ahead. Ultimately, success means visitors to our facility consistently receive accurate, up-to-date information that effectively promotes science and technology as DOST-STII's mission requires."

For Sir Allan:

"Can you walk us through a typical content update request from start to finish with the current system?"

"Currently, when my team needs to update content, they first create a document with the changes using Word or Excel, being very careful to specify exactly which AR marker needs updating. Then they email this to our technical **staff member who acts** as the intermediary. They forward it to the developers, who then have to schedule when

they can implement it. Sometimes they ask for clarification, which adds more email exchanges. Once they've made the change, they send us confirmation, but we can't verify it immediately - we must wait for them to deploy the update to the testing environment. If there's an issue, the whole process starts again. A simple change that should take minutes ends up taking 2-3 weeks from request to implementation, and during busy periods, it can take even longer."

"What features would make the CMS feel intuitive to you and your colleagues who aren't technically trained?"

"The most important feature would be a 'what you see is what you get' editor that shows exactly how the content will appear in the mobile app. If I can see the preview while editing, I won't have to guess about formatting. Simple drag-and-drop functionality for adding images would be helpful, rather than having to use complex upload processes. Clear visual indicators showing where content is in the approval process (like Encoder → Reviewer → Approver) would prevent confusion about status.

Having a 'quick edit' option for minor changes without going through the full workflow would save time. Most importantly, the system should use familiar interface patterns we already know from other websites - if it looks and works like tools, we use daily, the learning curve will be much smaller."

"How do you think having direct content control would impact visitor satisfaction with the AGHAMazing Quest experience?"

"Visitor satisfaction would improve significantly because we could ensure content is always current and accurate. Right now, when visitors point out errors or outdated information, we feel terrible knowing we can't fix it immediately. With direct control, we could correct mistakes the same day they're reported. We could also keep content fresh by regularly adding new information about recent scientific developments, which would make repeat visitors feel they're getting new value each time.

During special events or science observances, we could create timely content that connects with what's happening in the science world. Most importantly, visitors would receive consistent, up-to-date information that properly represents DOST-STII's work, which would enhance their overall experience and trust in our institution as a reliable source of science information."

Personas



ALAN C. TAULE

CHIEF, INFORMATION
RESOURCES AND ANALYSIS
DIVISION (IRAD)

CONTACT DETAILS



09239582384 / 09062540092



alan.taule@stii.dost.gov.ph



2nd floor, STII Bldg. DOST Complex,
General Santos Avenue, Bicutan,
Taguig City

ROLE

His role is tasked with curating, organizing, and analyzing STI resources to ensure their credibility, reliability, and accessibility. The AGHAMazing CMS complements this role by acting as a comprehensive knowledge management platform, equipped with metadata tagging, advanced search capabilities, and structured workflows that safeguard information integrity. It not only makes science and technology resources more discoverable but also provides avenues for disseminating data insights and reports in accessible formats. In doing so, the CMS strengthens IRAD's mission of transforming STI information into actionable knowledge for researchers, policymakers, and the public.



IRENE A. BRILLO

SCIENCE RESEARCH
SPECIALIST II (SRS)

CONTACT DETAILS



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STII Bldg. DOST Complex, General
Santos Avenue, Bicutan, Taguig City

NEEDS

- Centralized knowledge hub for resources
- Searchable and credible database of publications and reports
- Collaboration tools for research teams
- Version control to track research document changes
- Streamlined workflow for content approval and dissemination

WANTS

- Quick and seamless access to research outputs and resources
- Intuitive platform for uploading, storing, and retrieving documents
- Increased visibility for their research work
- Efficient peer-review and publication workflows

CRITERIA FOR SUCCESS

- Research outputs are accessible, accurate, and properly cited
- Smooth collaboration with colleagues and reviewers
- Timely approval and publication of outputs
- Increased visibility and dissemination of research findings
- A system that enhances productivity and minimizes admin burden

VALUES

- Accuracy, credibility, and accessibility of research data
- Effective communication and sharing of scientific knowledge
- Efficiency that saves time for actual research work
- Contribution of research to national development and innovation

FEARS

- Loss, misrepresentation, or delay of research outputs
- Limited visibility or recognition of their work
- Complex or hard-to-use CMS that hinders productivity
- System errors leading to incorrect dissemination of data



JONATHAN ABALON

TECHNICAL RESEARCH
SPECIALIST (TSR)

CONTACT DETAILS



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STII Bldg. DOST Complex, General
Santos Avenue, Bicutan, Taguig City

NEEDS

- Strong security (role-based access, authentication, encryption)
- Reliable performance and uptime
- Clear technical documentation for troubleshooting
- Backup and recovery mechanisms
- Integration with existing DOST systems

VALUES

- Efficiency and reliability in system operations
- Data integrity and sustainability
- Supporting staff to work effectively with minimal disruptions

CRITERIA FOR SUCCESS

- Minimal downtime and smooth system performance
- Strong compliance with security standards
- High adoption rates and fewer user complaints
- Seamless integration with other platforms
- Effective monitoring and support tools

WANTS

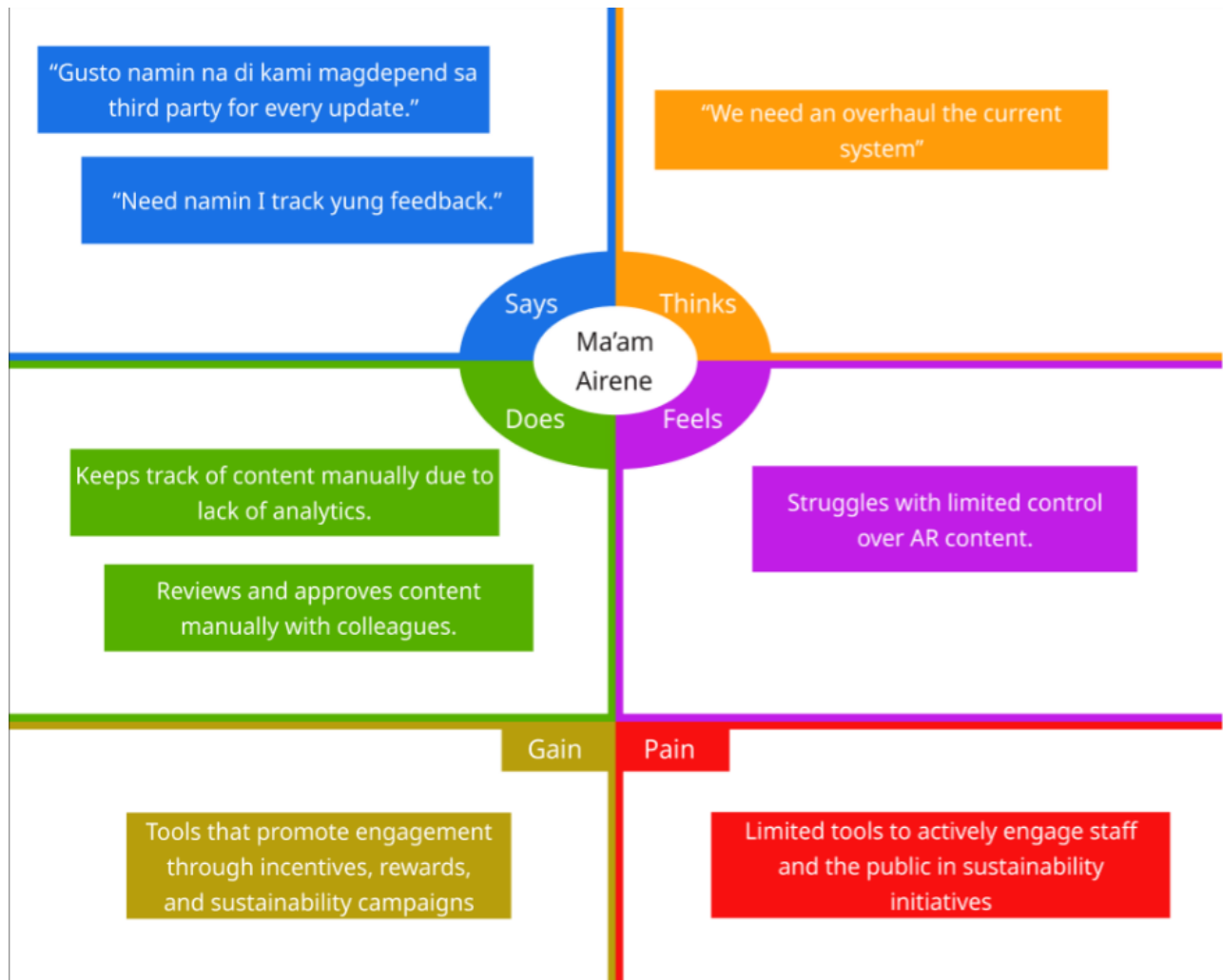
- Secure, stable, and easy-to-maintain CMS
- Minimal downtime and technical issues
- Tools for monitoring user activity and system performance
- Scalable system with easy updates

FEARS

- System crashes, cyberattacks, or data breaches
- Low adoption rates leading to constant helpdesk requests
- Lack of staff training causing improper system use
- Overcomplicated maintenance requirements

- Empathy Map







Pain-Gain Analysis Summary

Pains

- Dependence on developers for updating and maintaining content, which causes delays when urgent changes are required.
- Time-consuming and inefficient process of managing and publishing information, leading to outdated or inconsistent content
- Limited tools for staff/admins to keep content engaging, making it difficult to reach and maintain audience interest.
- Lack of centralized system, forcing staff to rely on scattered files and manual coordination.
- Inability to respond quickly to new scientific developments or correct errors promptly.
- Budget constraints due to ongoing developer costs for simple updates.
- Lack of centralized system, forcing staff to rely on scattered files and manual coordination.

Gains

- User-friendly CMS that enables staff/admins to manage and update content independently without technical expertise.
- Faster and more efficient workflows reduce reliance on developers and minimize bottlenecks.
- Features that allow easy formatting, scheduling, and organizing of content to keep platforms active and relevant.
- Centralized platform for managing content, ensuring consistency and accessibility for all admins.
- Role-based access control (Encoder, Reviewer, Approver) to maintain quality while enabling staff autonomy.
- Real-time content updates that improve visitor engagement and experience.

Customer Job

- Allow DOST staff and admins to manage and update content without technical barriers or developer intervention.
- Ensure information is consistently updated, accurate, and quickly accessible to the public.
- Improve communication and engagement with target audiences by making updates more interactive and timelier.
- Provide staff with tools that simplify their daily tasks and reduce manual workload.
- Maintain DOST branding and quality standards while enabling rapid content updates.
- Support DOST-STII's mission of promoting science and technology to the public through dynamic content

- **Stage 2 –Define**

Introduction

In the second stage of the design thinking process, we focused on the gathered information from Stage 1 (Empathize), where we interviewed DOST-STII staff including Ms. Irene Brillo, Sir Jonathan Abalon, and Ms. Grace Ann Valondo. We analyzed the pain points, needs, and insights of the users to articulate a clear Point of View (POV) that would help us form our problem statement.

Our team reviewed the interview transcripts, observations, and pain-gain analysis from Stage 1 to identify key issues. We organized these insights using affinity mapping to cluster related problems and identify patterns.

Key Issues Identified:

- DOST-STII staff are fully dependent on third-party developers for content updates to the AGHAMazing Quest mobile application
- Content updates require lengthy coordination with developers, causing delays
- The current system doesn't allow DOST-STII staff to make immediate changes when needed
- Staff feel they lack ownership of the application they use daily
- The inability to quickly update content affects DOST-STII's mission to promote science and technology to the public
- There is a need for quality control through a structured workflow (Encoder → Reviewer → Approver)

Problem Statement Development

After analyzing our empathy findings, we crafted the following problem statement:

"How might we create a user-friendly Content Management System that enables DOST-STII staff to independently manage and update content for the AGHAMazing Quest mobile application, reducing dependency on developers while maintaining the integrity of the application?"

How Might We Questions:

Based on our problem statement, we developed the following "How Might We" questions to guide our ideation phase:

1. How might we design a CMS interface that requires no technical expertise for DOST-STII staff to use?
2. How might we implement a content workflow system that supports the Encoder, Reviewer, and Approver roles requested by DOST-STII?
3. How might we create a system that allows for immediate content updates while ensuring content quality?
4. How might we integrate the CMS with the existing AGHAMazing Quest mobile application seamlessly?
5. How might we design a CMS that aligns with DOST branding guidelines while maintaining a modern, innovative appearance?
6. How might we provide DOST-STII staff with analytics to understand how visitors interact with the content?
7. How might we ensure the CMS is intuitive enough for non-technical staff to adopt quickly?
8. How might we create a preview feature that shows exactly how content will appear in the mobile app?
9. How might we implement version control to allow staff to revert to previous content if needed?

10. How might we structure the CMS to accommodate future expansion of features and content types?

- **Stage 3 – Ideate**

Introduction

In the third stage of the design thinking process, we focused on generating innovative solutions to address the problem statement developed in Stage 2. Our goal was to brainstorm a wide range of potential features and approaches for the AGHAMazing Quest CMS that would meet DOST-STII's needs while being feasible to implement.

Brainstorming session

Our team conducted a structured brainstorming session using the "How Might We" questions from Stage 2 as prompts. We generated over 40 potential features and solutions, which we then organized into categories:

1. Content Management Features
 - a. Rich text editor with image/video embedding
 - b. Version history and rollback capability
 - c. Content scheduling for future updates
 - d. Draft/publish workflow with approval system
 - e. WYSIWYG editor with mobile app preview
2. User Management Features
 - a. Role-based access control (Encoder, Reviewer, Approver)
 - b. User activity tracking
 - c. Simple onboarding for new staff members
 - d. Customizable user dashboards
3. Integration Features
 - a. API connection to mobile application
 - b. Real-time content synchronization
 - c. AR marker management system
 - d. Two-way sync between CMS and mobile app
4. Analytics Features
 - a. Content engagement metrics
 - b. Visitor interaction tracking
 - c. Popular content identification

d. Content performance reporting

Crazy 8's

Our team conducted a Crazy 8's exercise where each member sketched 8 different interface concepts in 8 minutes. The sketches focused on:

- Dashboard layout for content management
- Content editing interface
- User role management screens
- Workflow approval process visualization
- Analytics display formats
- Mobile-responsive design approaches
- AR marker management interfaces
- Content preview functionality

Post-it voting

After the Crazy 8's exercise, we reviewed all sketches and placed sticky notes on features we believed were most promising. The top-voted concepts were:

1. Dashboard with content status overview (showing draft, approved, published content)
2. Intuitive drag-and-drop content editor with real-time mobile app preview
3. Visual workflow system showing content through Encoder → Reviewer → Approver process
4. Simple media library with search functionality and AR marker association
5. Mobile-responsive design that works on various devices
6. Version history timeline with easy rollback capability
7. Content scheduling calendar for timed updates
8. Analytics dashboard showing visitor engagement metrics

Desirable, viable, feasible session

After selecting our best sketches, we now tried to utilize its desirable, viable, and feasible application on our end. We have discussed and voted on each panel per member, and the following results are: We evaluated our top concepts using the DVF framework:

1. Dashboard with content status overview

- Desirability: High (staff need visibility into content status)
- Viability: High (aligns with DOST-STII's workflow needs)
- Feasibility: High (standard CMS feature that can be implemented)

2. Intuitive drag-and-drop content editor with preview

- Desirability: High (reduces technical barriers for staff)
- Viability: High (essential for non-technical users)
- Feasibility: High (many open-source components available)

3. Visual workflow system

- Desirability: High (matches requested roles: Encoder, Reviewer, Approver)
- Viability: High (core requirement from client)
- Feasibility: Medium (requires careful implementation but doable)

4. Media library with AR marker association

- Desirability: High (staff need to manage AR content efficiently)
- Viability: High (necessary for content management)
- Feasibility: High (standard feature in CMS platforms)

5. Mobile-responsive design

- Desirability: Medium (staff primarily use desktops but some mobility needed)
- Viability: Medium (not critical but helpful)

- Feasibility: High (responsive design is standard practice)

6. Version history timeline

- Desirability: High (staff need to correct mistakes without developer help)
- Viability: High (critical for content quality control)
- Feasibility: High (common CMS feature)

7. Content scheduling calendar

- Desirability: Medium (useful for planned updates)
- Viability: Medium (nice-to-have but not essential)
- Feasibility: High (standard feature)

8. Analytics dashboard

- Desirability: Medium (helps understand content effectiveness)
- Viability: Medium (supports decision making)
- Feasibility: Medium (requires integration with mobile app)

• Stage 4 – Prototype

Introduction

In the fourth stage of the design thinking process, we, Eco-Coders, designed prototype solutions for the AGHAMazing Quest CMS. Based on our ideation phase, we created wireframes for a web-based CMS that addresses DOST-STII's needs for content management, workflow, and independence from developer support.

Our prototype focuses on the core features identified through our design thinking process, with special attention to the user roles (Encoder, Reviewer, Approver) requested by DOST-STII and the need for a modern, intuitive interface.

Wireframe Designs

We created low-fidelity wireframes for key CMS functionality:

1. Login Screen

- Simple local authentication interface with DOST branding
- Secure login process for different user roles
- Password recovery functionality

2. Dashboard Overview

- Shows content status (draft, review, approved, published)
- Quick access to most used functions
- Visual indicators for pending approvals
- Content update statistics and analytics snapshot

3. Content Editor Interface

- Rich text editor with image/video embedding
- Preview functionality showing how content appears in mobile app
- Version history tracking with comparison tool
- AR marker association selector

4. Workflow Management

process

- Visual representation of content through Encoder → Reviewer → Approver

- Notification system for pending actions
- Comments and feedback capability between roles
- Approval/rejection with reason functionality

5. Media Management

- Organized library for images, videos, and documents
- Search functionality for quick content retrieval
- Integration with AR markers used in the mobile app
- Drag-and-drop upload capability

6. User Management

- Interface for managing user accounts and roles
- Activity tracking for accountability
- Simple onboarding process for new staff

7. Analytics Dashboard

- Basic reporting on content usage
- Visitor engagement metrics
- Popular content identification
- Content performance trends

Discussion

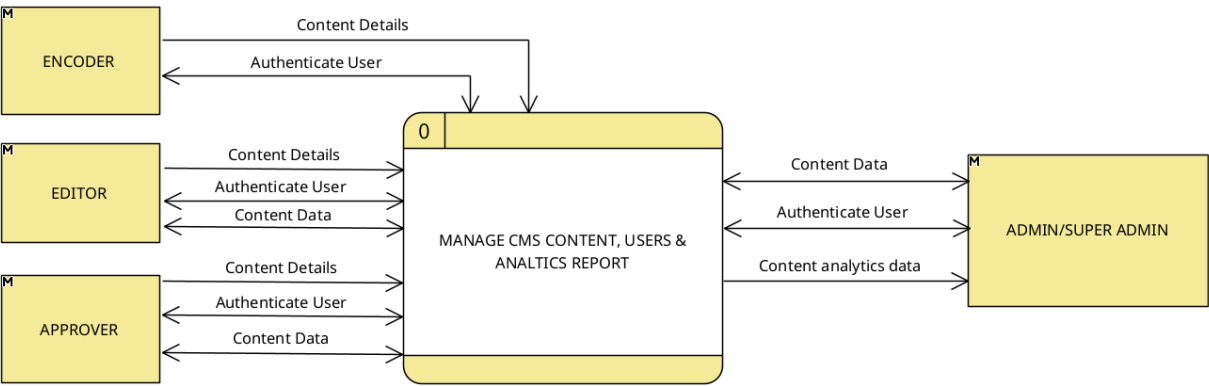
Mockups and sketching



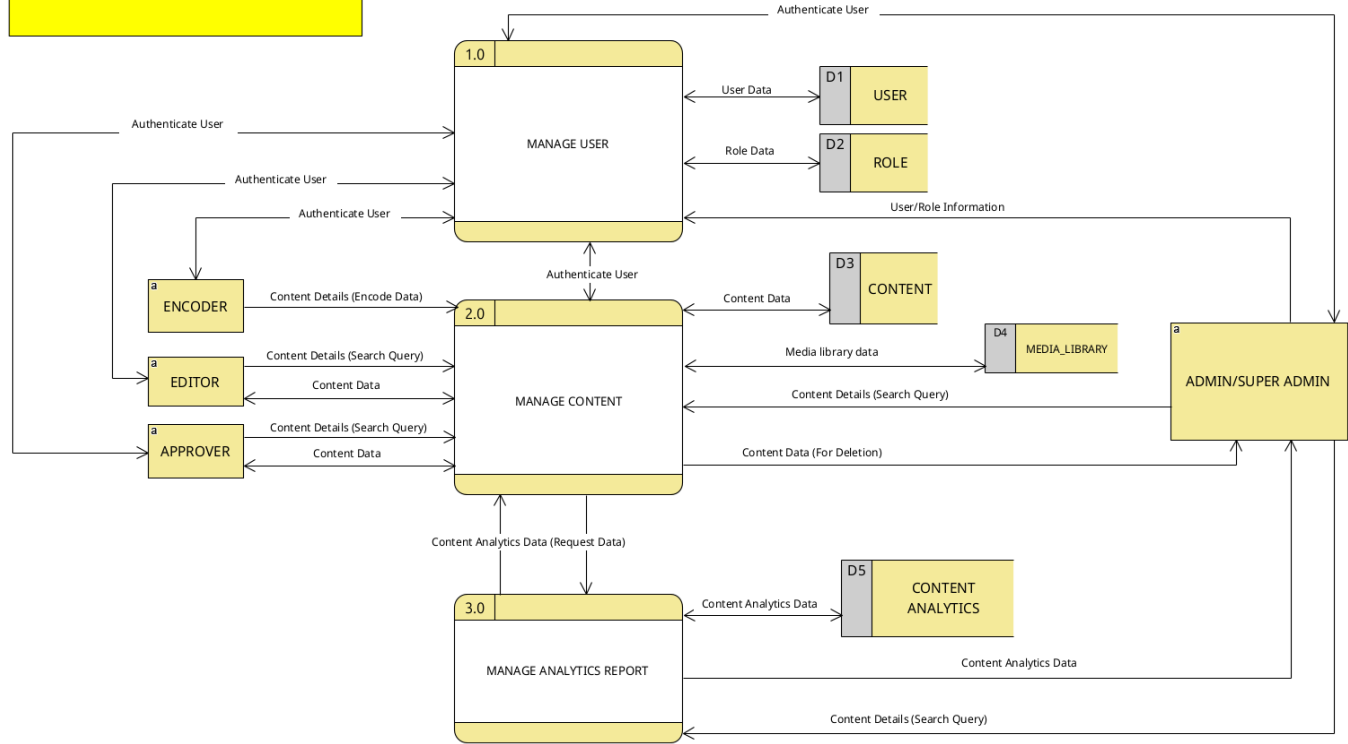
- Stage 5 - Test

Dataflow Diagrams

DFD LEVEL 0

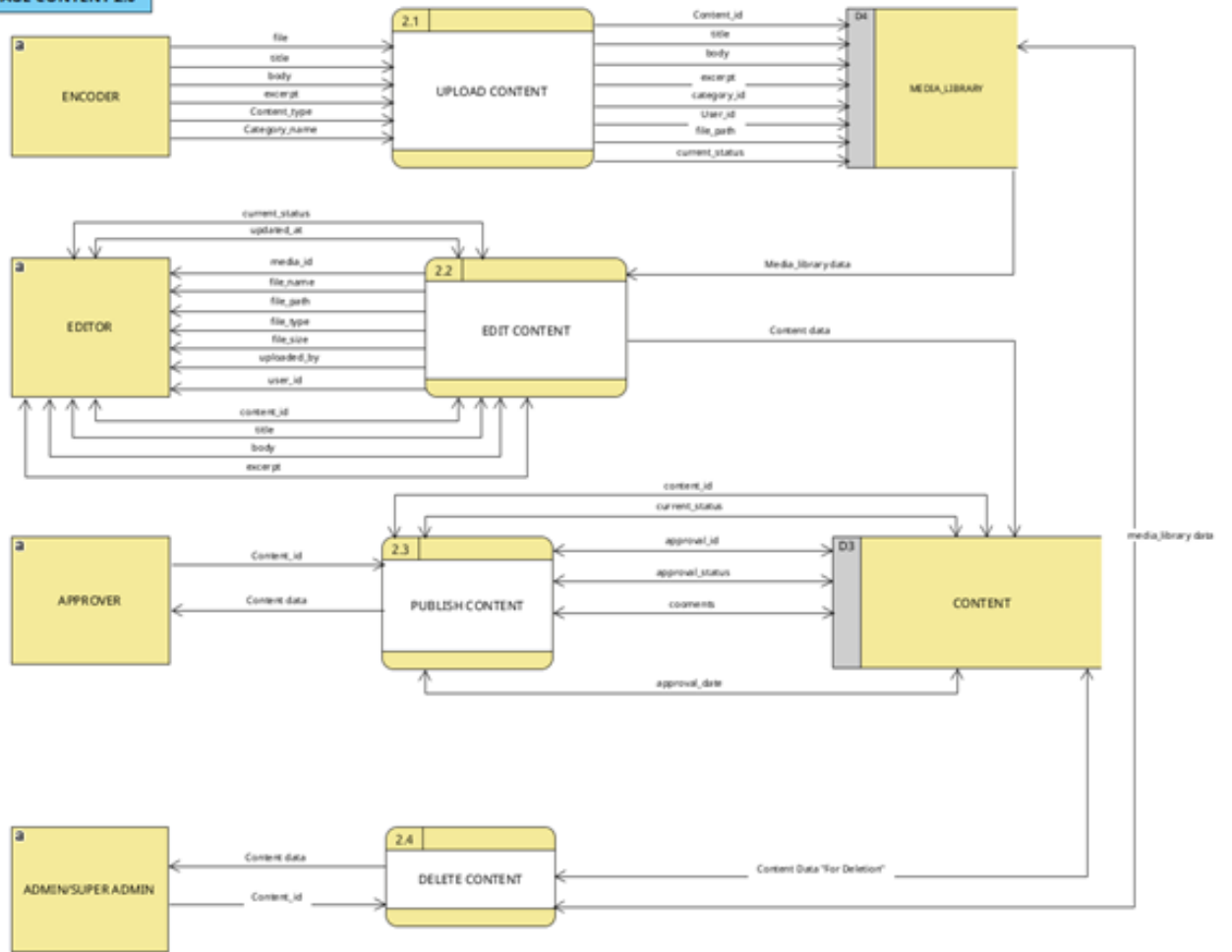


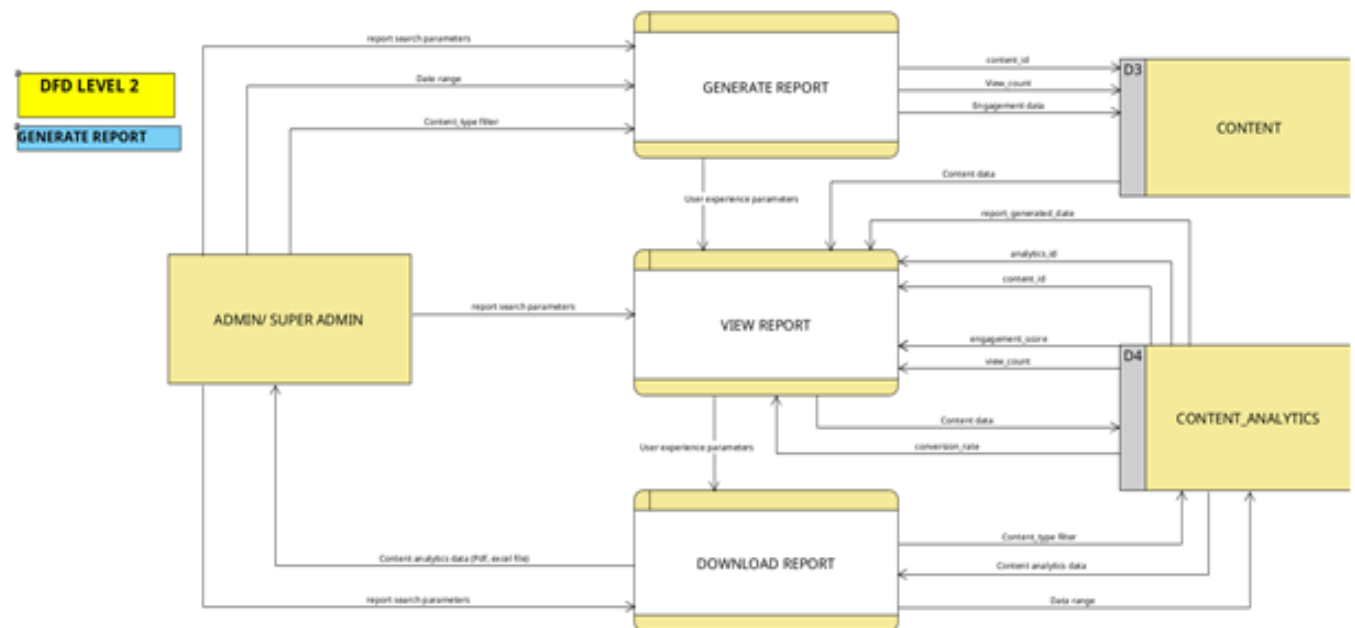
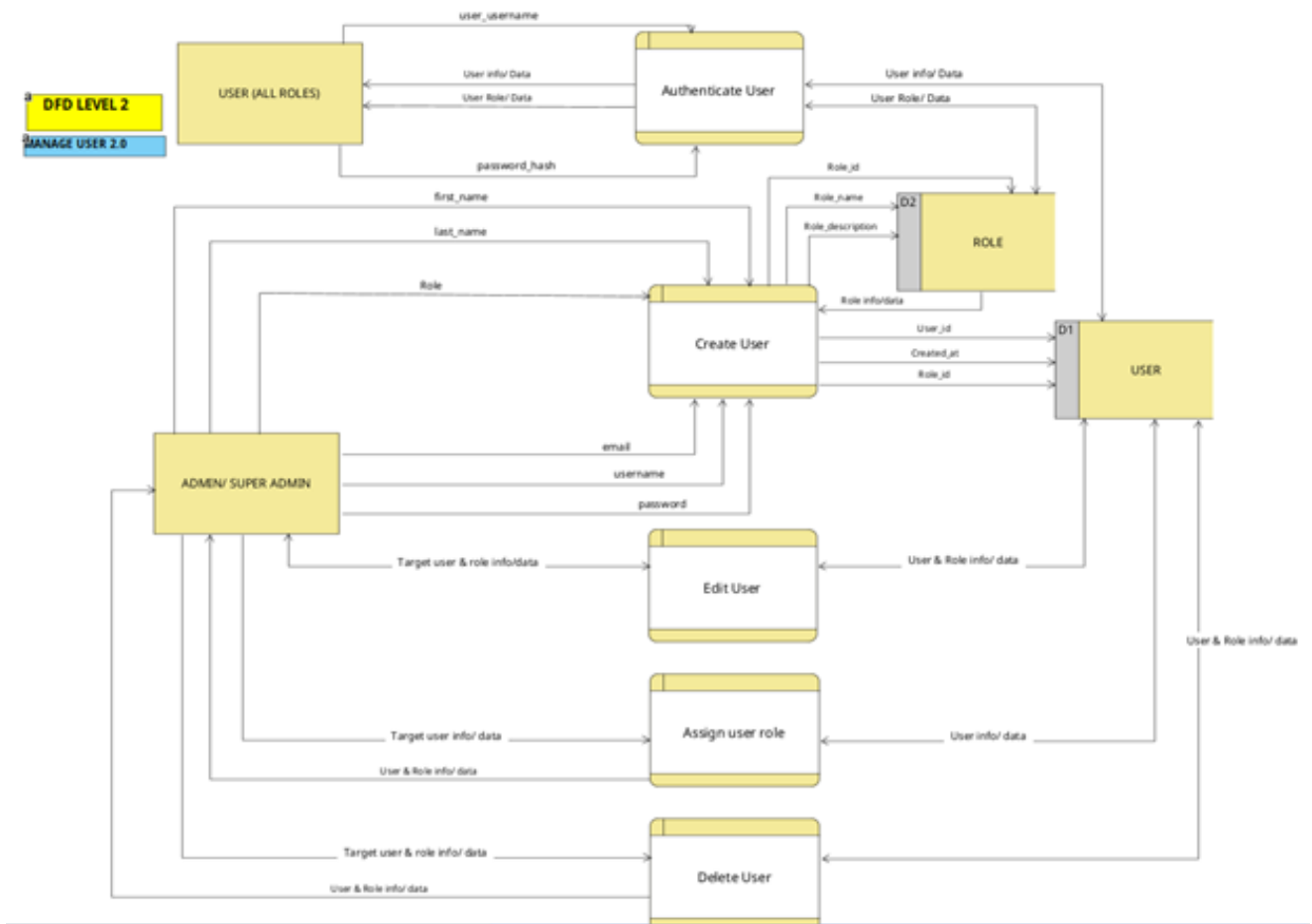
DFD LEVEL 1 CMS CORE PROCESS



DFD LEVEL 2

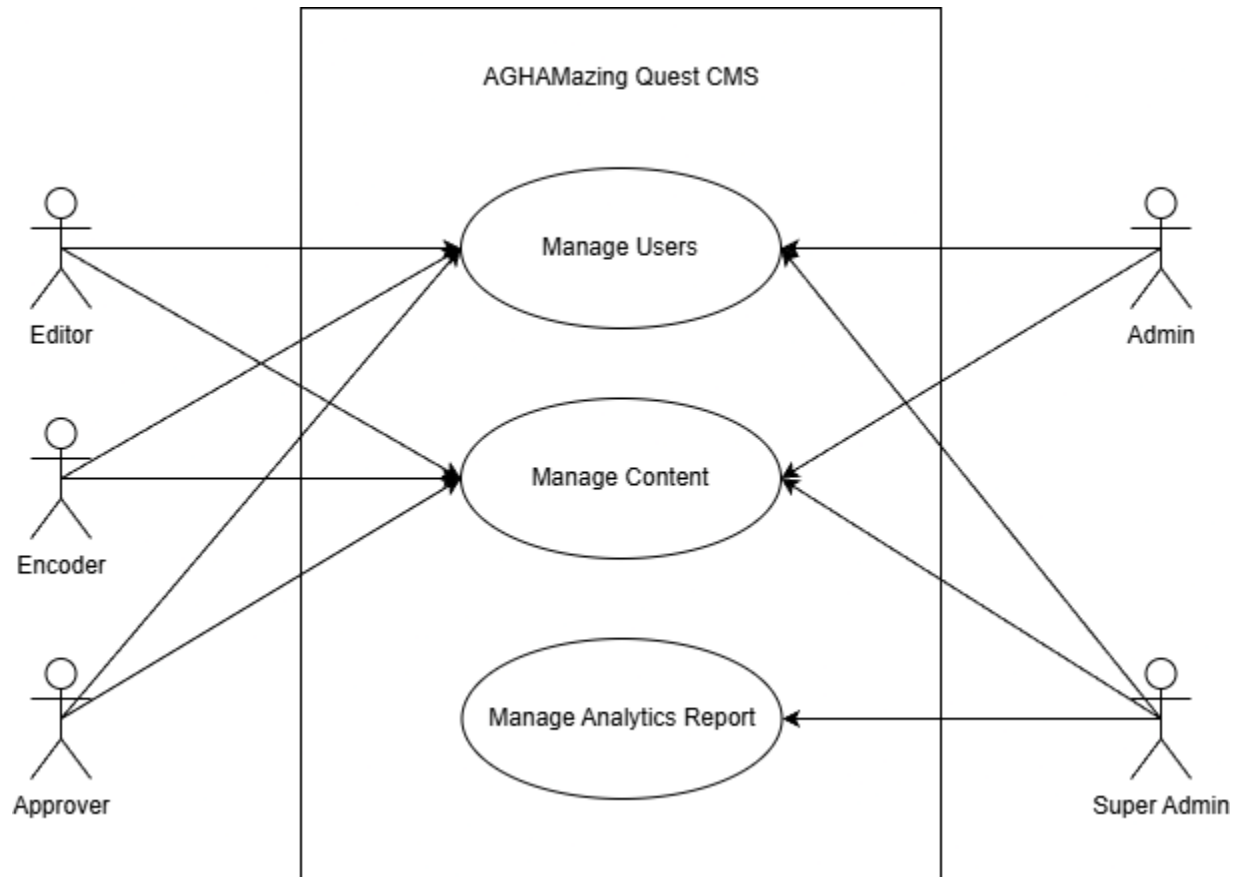
MANAGE CONTENT 2.0





Use Case Documentations

1.0 Use Case Diagram



2.0 Use Case Fully Dressed

Use Case 1.0 Manage Users

Use Case Name	Authentication
Use Case ID	UC1.1
Author	Ervyne Anthony V. Chan
Purpose	To verify user credentials and prevent unauthorized access.
Requirement Traceability	
Priority	High

Preconditions	1. The user must have a registered account.
Postconditions	1. The user is successfully logged into the system and granted access based on their role.
Actors	Encoder, Editor, Approver, Admin, Super Admin
Basic Flow	1. The user navigates to the login screen. 2. The user enters their username and password. 3. The system validates the credentials. 4. If valid, the system logs the user in and redirects to the dashboard. 5. The system displays a success message: "Login successful."
Alternate Flow	1. The user navigates to the login screen. 2. The user enters their username and password. 3. The system validates the credentials. 4. If invalid, the system displays error message 5. The system returns you to the login page.

Use Case Name	Create User
Use Case ID	UC1.2
Author	Ervyne Anthony V. Chan
Purpose	To allow Admins or Super Admins to create users.
Requirement Traceability	
Priority	High

Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. The created user will be given a “User ID”
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “User Management” section. 2. The user clicks the “Create User” button. 3. The system displays a form for user details. 4. The user fills in the required fields and submits the form. 5. The system validates the input and creates the user account. 6. The system displays a success message: “User successfully created.”
Alternate Flow	1. The user navigates to the “User Management” section. 2. The user clicks the “Create User” button. 3. The system displays a form for user details. 4. The user fills in the required fields and submits the form. 5. The system validates the input. 6. The system detects that there is an existing user. 7. The system returns the user to fill up required fields.

Use Case Name	Edit User
Use Case ID	UC1.3
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to edit an existing user’s details.

Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “User Management” section. 2. The user selects a user from the list. 3. The system displays the selected user’s details. 4. The user clicks the “Edit” button. 5. The system displays an editable form. 6. The user modifies the necessary fields and submits the form. 7. The system validates the input and updates the user record. 8. The system displays a success message: “User details successfully updated.”
Alternate Flow	1. The user navigates to the “User Management” section. 2. The user selects a user from the list. 3. The system displays the selected user’s details. 4. The user clicks the “Edit” button. 5. The system displays an editable form. 6. The user modifies the necessary fields and submits the form. 7. The system validates the input. (Invalid)

	8. The system displays errors 9. The system returns you to edit the form for resubmission.
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Use Case Name	Create User Role
Use Case ID	UC1.4
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to create user roles with specific permissions.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. A new user role is created and stored in the database. 2. The role can be assigned to users.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “Role Management” section. 2. The user clicks the “Create Role” button. 3. The system displays a form for role details and permissions. 4. The user fills in the required fields and submits the form. 5. The system validates the input and creates the role. 6. The system displays a success message: “Role successfully created.”

Alternate Flow	1. The user navigates to the “Role Management” section. 2. The user clicks the “Create Role” button. 3. The system displays a form for role details and permissions. 4. The user fills in the required fields and submits the form. 5. The system validates the input and creates the role. (Invalid) 6. The system displays errors and returns the user to answer form.
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Use Case Name	Edit User Role
Use Case ID	UC1.5
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to edit an existing user role and its permissions.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. The selected role is updated in the database. 2. Changes are reflected in the permissions of users assigned to that role.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “Role Management” section.

	2. The user selects a role from the list. 3. The system displays the role's details. 4. The user clicks the "Edit" button. 5. The system displays an editable form. 6. The user modifies the necessary fields and submits the form. 7. The system validates the input and updates the role. (Yes) 8. The system displays a success message: "Role successfully updated."
Alternate Flow	1. The user navigates to the "Role Management" section. 2. The user selects a role from the list. 3. The system displays the role's details. 4. The user clicks the "Edit" button. 5. The system displays an editable form. 6. The user modifies the necessary fields and submits the form. 7. The system validates the input. (Invalid) 8. The system displays errors

Use Case Name	Assign User Role
Use Case ID	UC1.6
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to assign users with a specific role.

Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role. 3. The target user must be registered in the system.
Postconditions	1. The selected role is assigned to the user. 2. The user’s access and permissions are updated accordingly.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “User Management” section. 2. The user selects a user from the list. 3. The system displays the user’s details. 4. The user clicks the “Assign Role” button. 5. The system displays a list of available roles. 6. The user selects a role and confirms the assignment. 7. The system validates the role assignment. (Valid) 8. The system updates the user’s role and displays a success message: “Role successfully assigned.”
Alternate Flow	1. The user navigates to the “User Management” section. 2. The user selects a user from the list. 3. The system displays the user’s details. 4. The user clicks the “Assign Role” button. 5. The system displays a list of available roles.

	6. The user selects a role and confirms the assignment. 7. The system validates the role assignment. (Invalid) 8. The system displays error message “Role cannot be given to user”
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Use Case Name	Delete User
Use Case ID	UC1.7
Author	Ervyne Anthony V. Chan
Purpose	To allow Admins or Super Admins to delete users.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. The selected user account is deleted from the database. 2. The user can no longer log in or access the system.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “User Management” section. 2. The user selects a user from the list. 3. The system displays the user’s details. 4. The user clicks the “Delete” button. 5. The system prompts confirmation: “Are you sure you want to delete this user?”

	- The user confirms the deletion. - The system deletes the user account from the database. - The system displays a success message: "User successfully deleted."
Alternate Flow	- The user navigates to the "User Management" section. - The user selects a user from the list. - The system displays the user's details. - The user clicks the "Delete" button. - The system prompts confirmation: "Are you sure you want to delete this user?" - The user cancels the deletion. - The system returns the user to the user list.

Use Case 2.0 Manage Content

Use Case Name	Upload Content
Use Case ID	UC2.1
Author	Ervyne Anthony V. Chan
Purpose	To be able to upload content
Requirement Traceability	
Priority	High

Preconditions	1. The user must be logged into the system. 2. The user must have “Encoder” or “Editor” role.
Postconditions	1. The uploaded content will be given a “Content ID”. 2. The uploaded content will be given the status “For Final Editing”. 3. The uploaded content will be saved in the database.
Actors	Encoder, Editor
Basic Flow	1. The Encoder or Editor will click the “Upload Content” button. 2. The Encoder or Editor will select a file from their device. 3. The Encoder or Editor will click the “Upload” button. 4. The system will confirm the upload and output “Content Uploaded”. 5. The system will provide a “Content ID” to the uploaded content. 6. The system will set the status of the uploaded content as “For Final Editing” 7. The system will save the uploaded content into the database.
Alternate Flow	1. The Encoder or Editor will click the “Upload Content” button. 2. The Encoder or Editor will select a file from their device. 3. The Encoder or Editor will click the “Upload” button. 4. The system will confirm that the upload has failed and will output “Content Failed to Upload”. 5. The system will request the user to upload another file.

Use Case Name	Edit Content
Use Case ID	UC2.2
Author	Ervyne Anthony V. Chan
Purpose	To be able to edit already uploaded content.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have “Editor” role.
Postconditions	1. The edited content will be given the status “For Approval”.
Actors	Editor
Basic Flow	1. The Editor will click the “Search” button. 2. The Editor will click the “Content ID” filter. 3. The Editor will input the “Content ID”. 4. The system will check if content is found. (Yes) 5. The system will output the specific uploaded content with the searched “Content ID”. 6. The Editor will conduct the needed changes on the already uploaded content. 7. The Editor will click the “Save Changes” button. 8. The system will set the status of the edited content as “For Approval”.

	9. The system will save the edited content into the database.
Alternate Flow	1. The Editor will click the “Search” button. 2. The Editor will click the “Content ID” filter. 3. The Editor will input the “Content ID”. 4. The system will check if content is found. (No) 5. The system will display “Content not Found”

Use Case Name	Approve Content
Use Case ID	UC2.3
Author	Ervyne Anthony V. Chan
Purpose	To be able to approve content for publishing.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have the “Approver” role. 3. The content status should be “For Approval”.
Postconditions	1. The approved content will be given the status “For Publishing”
Actors	Approver
Basic Flow	1. The Approver will click the “Search” button. 2. The Approver will click the “For Approval” filter. 3. The Approver will input a “Content ID”. 4. The system will search for content with “For Approval” status.

	5. The system has content with “For Approval” status. 6. The system will output the content list with “For Approval” status. 7. The Approver will choose from the list of content. 8. The system will output the chosen content. 9. The Approver will verify the content. 10. The Approver will click the “Approve” button. 11. The system will set the status of the approved content to "For Publishing". 12. The system will save the status update to the database.
Alternate Flow	1. The Approver will click the “Search” button. 2. The Approver will click the “For Approval” filter. 3. The Approver will input a “Content ID”. 4. The system will search for content with “For Approval” status. 5. The system has no content with “For Approval” status. 6. Display “No content available for Approval”
Alternate Flow	1. The Approver will click the “Search” button. 2. The Approver will click the “For Approval” filter. 3. The Approver will input a “Content ID”. 4. The system will search for content with “For Approval” status. 5. The system has content with “For Approval” status. 6. The system will output the content list with “For Approval” status.

	7. The Approver will choose from the list of content. 8. The system will output the chosen content. 9. The Approver will verify the content. 10. The Approver will click the “Disapprove” button. 11. The system will set the status of the disapproved content to “For editing”. 12. The system will save the status update to the database.
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Use Case Name	Publish Content
Use Case ID	UC2.4
Author	Ervyne Anthony V. Chan
Purpose	To be able to publish approved content.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have the “Approver” role. 3. The content to be published must have the “For Publishing” status.
Postconditions	1. The content published will be given the status “Published”. 2. The published content will be saved into the database where the mobile application can be accessed.
Actors	Approver

Basic Flow	1. The Approver clicks the “Search” button. 2. The Approver selects the “For Publishing” filter. 3. The system displays content items with the status “For Publishing”. 4. The Approver selects a content item from the list. 5. The system displays the selected content. 6. The Approver reviews the content for final publishing. 7. The Approver clicks the “Publish” button. 8. The system updates the content status to “Published”. 9. The system saves the published content to the database. 10. The content becomes accessible to the mobile application.
Alternate Flow	1. The Approver clicks the “Search” button. 2. The Approver selects the “For Publishing” filter. 3. The system searches database for content with “For publishing” status. 4. The system has no content found 5. The system displays “None ready for publishing”

Use Case Name	Delete Content
Use Case ID	UC2.5
Author	Ervyne Anthony V. Chan
Purpose	To be able to delete content.

Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. The selected content will be deleted from the database. 2. The content will no longer be accessible via the CMS or mobile application.
Actors	Admin, Super Admin
Basic Flow	1. The user will click the “Delete Content” button. 2. The system will show the list of content within the database. 3. The user will select the specific content item. 4. The system will show the selected content. 5. The user will click the “Delete” button. 6. The system prompts confirmation “Are you sure you want to delete this content?”. 7. The user confirms the deletion. 8. The system deletes the content from the database. 9. The system displays a success message: “Content successfully deleted.”
Alternate Flow	1. The user will click the “Delete Content” button. 2. The system will show the list of content within the database. 3. The user will select the specific content item.

	4. The system will show the selected content. 5. The user will click the “Delete” button. 6. The system prompts confirmation “Are you sure you want to delete this content?”. 7. The user declines deletion. 8. The system cancels deletion. 9. The system returns the user to the content list.
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Use Case 3.0 Manage Analytics Reports

Use Case Name	Generate Analytics Reports
Use Case ID	UC3.1
Author	Ervyne Anthony V. Chan
Purpose	To allow Admin or Super Admin to generate analytics reports based on system data.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role.
Postconditions	1. A new analytics report is generated and stored in the system.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “Analytics Reports” section. 2. The system displays a list of analytics.

	3. The user clicks the “Generate Report” button. 4. The system displays a form to select report metrics. 5. The user fills in the required fields and submits the form. 6. The system processes the data and generates the report. 7. The system displays a success message: “Report successfully generated.”
Alternate Flow	1. The user navigates to the “Analytics Reports” section. 2. The system displays a list of analytics. 3. The user clicks the “Generate Report” button. 4. The system displays a form to select report metrics. 5. The user fills in the required fields and submits the form. 6. The system validates input. (Invalid) 7. The system rejects the form and asks the user to resubmit. 8. The system returns you to fill out the form.
Alternate Flow	1. The user navigates to the “Analytics Reports” section. 2. The system displays a list of analytics. 3. The user clicks the “Generate Report” button. 4. The system displays a form to select report metrics. 5. The user fills in the required fields and submits the form. 6. The system validates input. (Duplicate Report) 7. The system rejects the form and displays that the report exists already.

Use Case Name	View Analytics Reports
Use Case ID	UC3.2
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to view previously generated analytics reports.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role. 3. At least one report must be saved in the system.
Postconditions	1. The selected report is displayed on the screen.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “Analytics Reports” section. 2. The system checks if there are reports available. (Yes) 3. The system displays a list of available reports. 4. The user selects a report from the list. 5. The system displays the report's content.
Alternate Flow	1. The user navigates to the “Analytics Reports” section. 2. The system checks if there are reports available. (No) 3. The system displays “No reports have been made”.

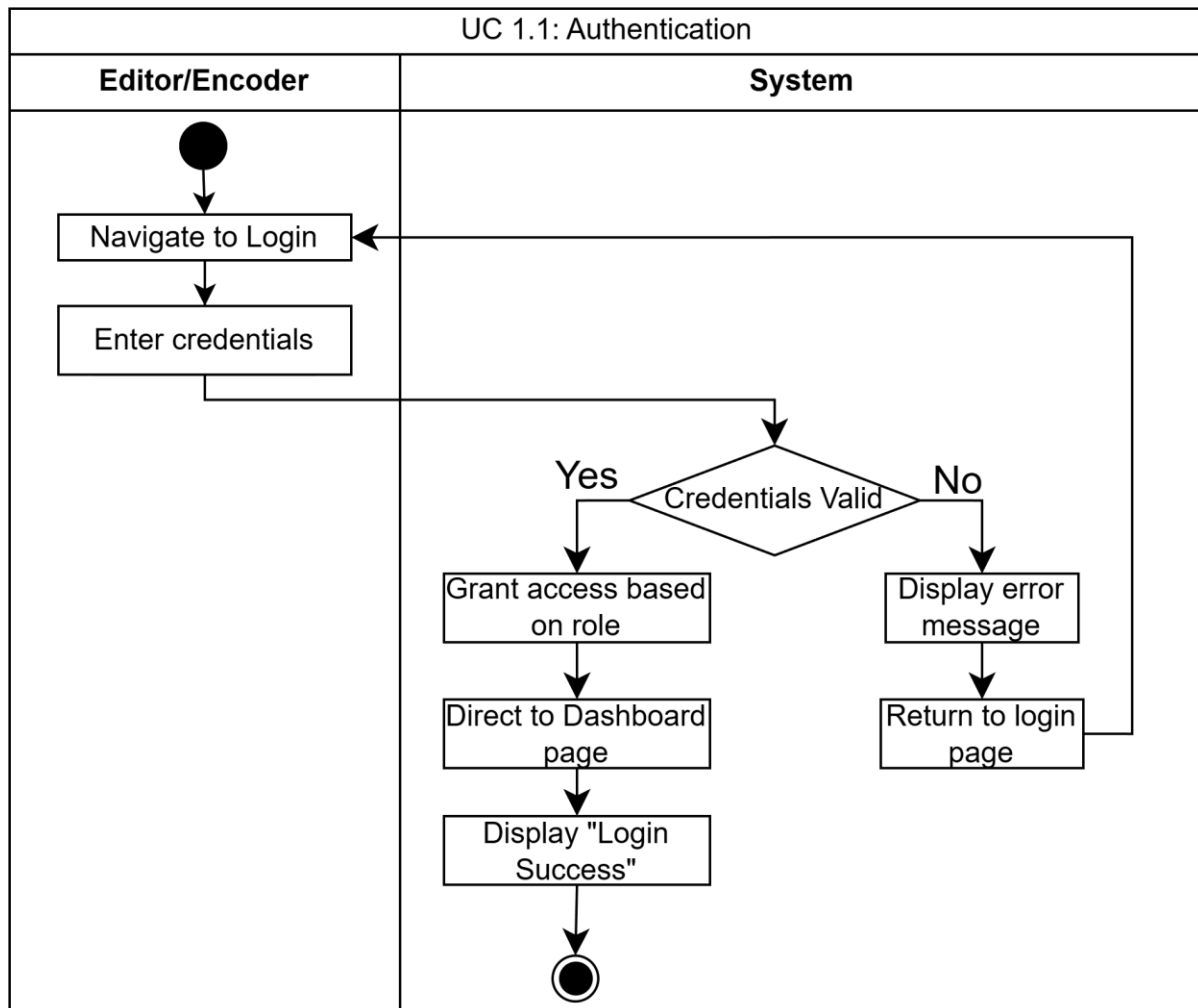
	4. The system returns the user to the Analytics Reports section.
--	--

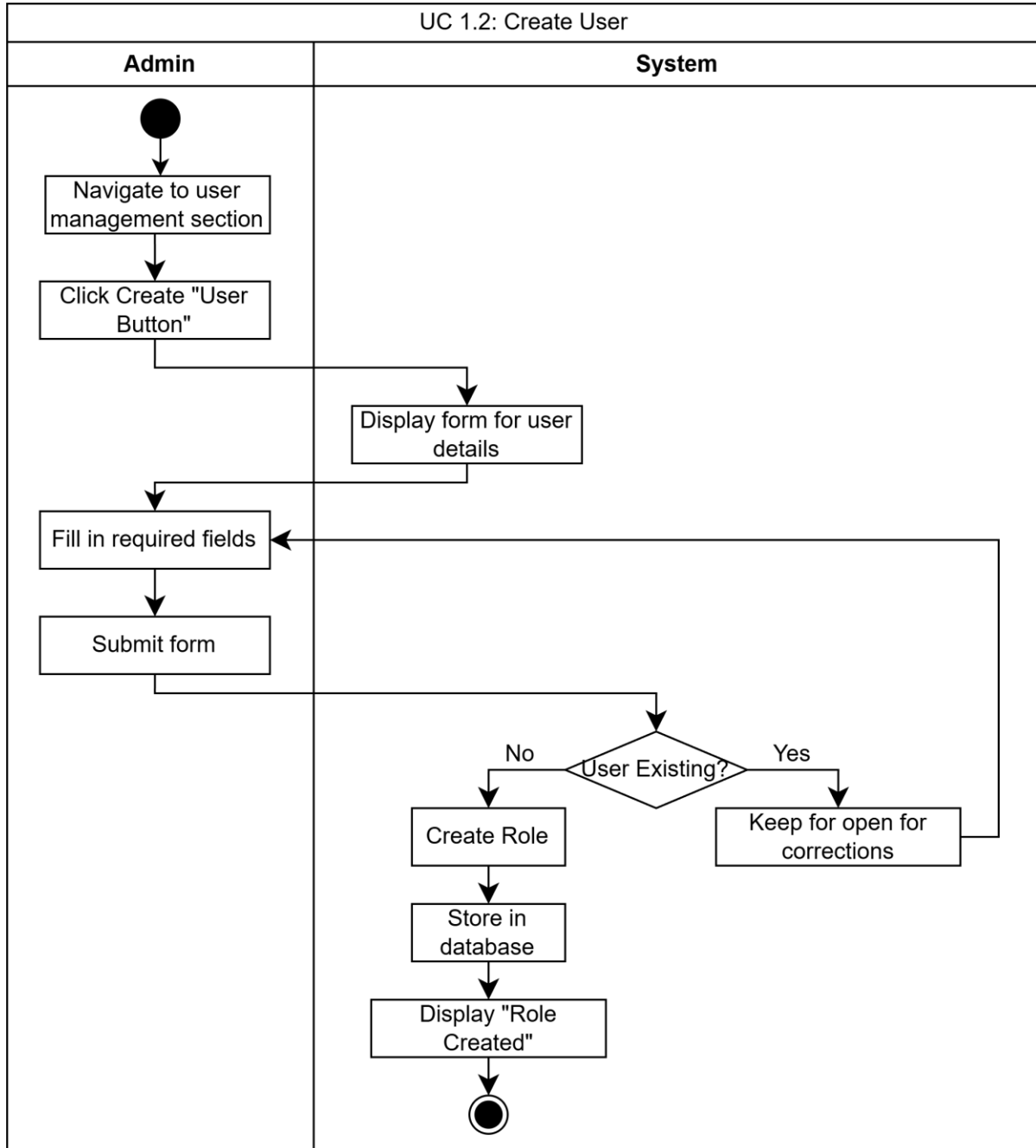
Use Case Name	Download Analytics Reports
Use Case ID	UC3.3
Author	Ervyne Anthony V. Chan
Purpose	To allow an Admin or Super Admin to download analytics reports for offline use or sharing.
Requirement Traceability	
Priority	High
Preconditions	1. The user must be logged into the system. 2. The user must have either an “Admin” or “Super Admin” role. 3. At least one report must exist in the system.
Postconditions	1. The selected report is downloaded to the user’s device.
Actors	Admin, Super Admin
Basic Flow	1. The user navigates to the “Analytics Reports” section. 2. The system checks if there are reports available. (Yes) 3. The system displays a list of available reports. 4. The user selects a report 5. The system makes sure it's the correct report. (Yes) 6. The system prepares the report file.

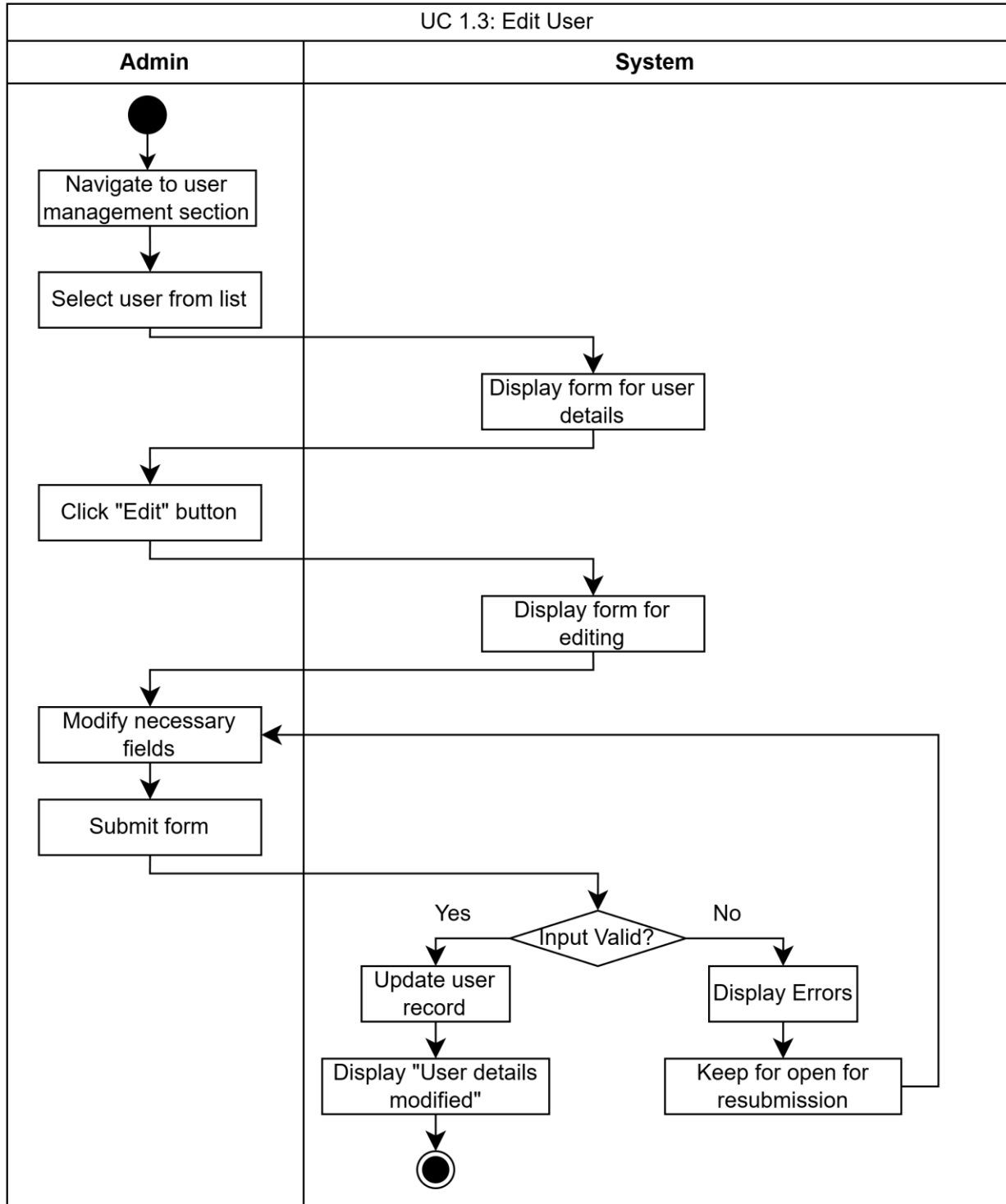
	7. The system initiates the download and displays a success message: "Report successfully downloaded."
Alternate Flow	1. The user navigates to the "Analytics Reports" section. 2. The system checks if there are reports available. (No) 3. The system displays "No reports have been made". 4. The system returns the user to the Analytics Reports section.
Alternate Flow	1. The user navigates to the "Analytics Reports" section. 2. The system checks if there are reports available. (Yes) 3. The system displays a list of available reports. 4. The user selects a report 5. The system makes sure it's the correct report. (No) 6. The system returns the user to select a report.

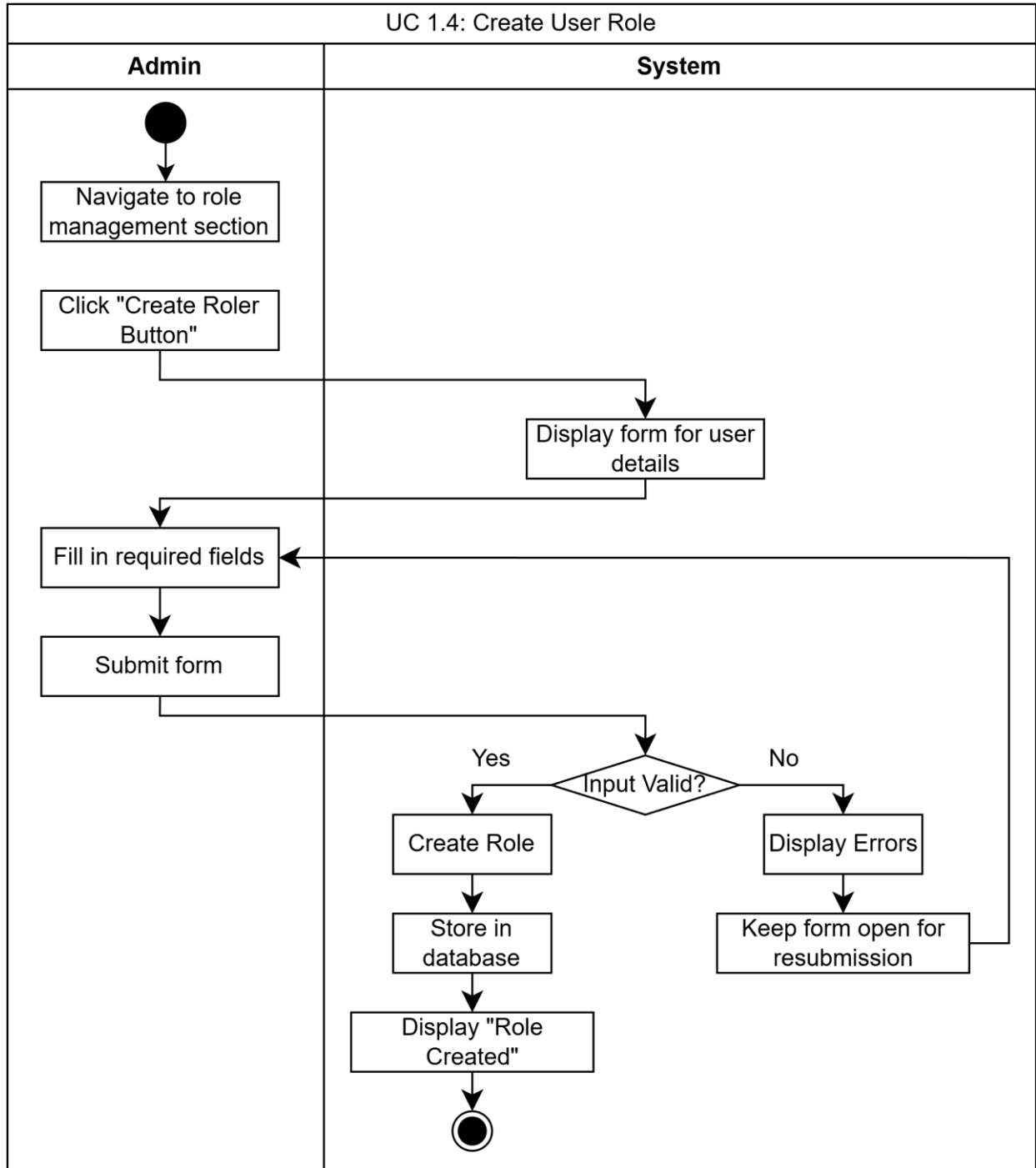
3.0 Activity Diagrams with Swim Lanes

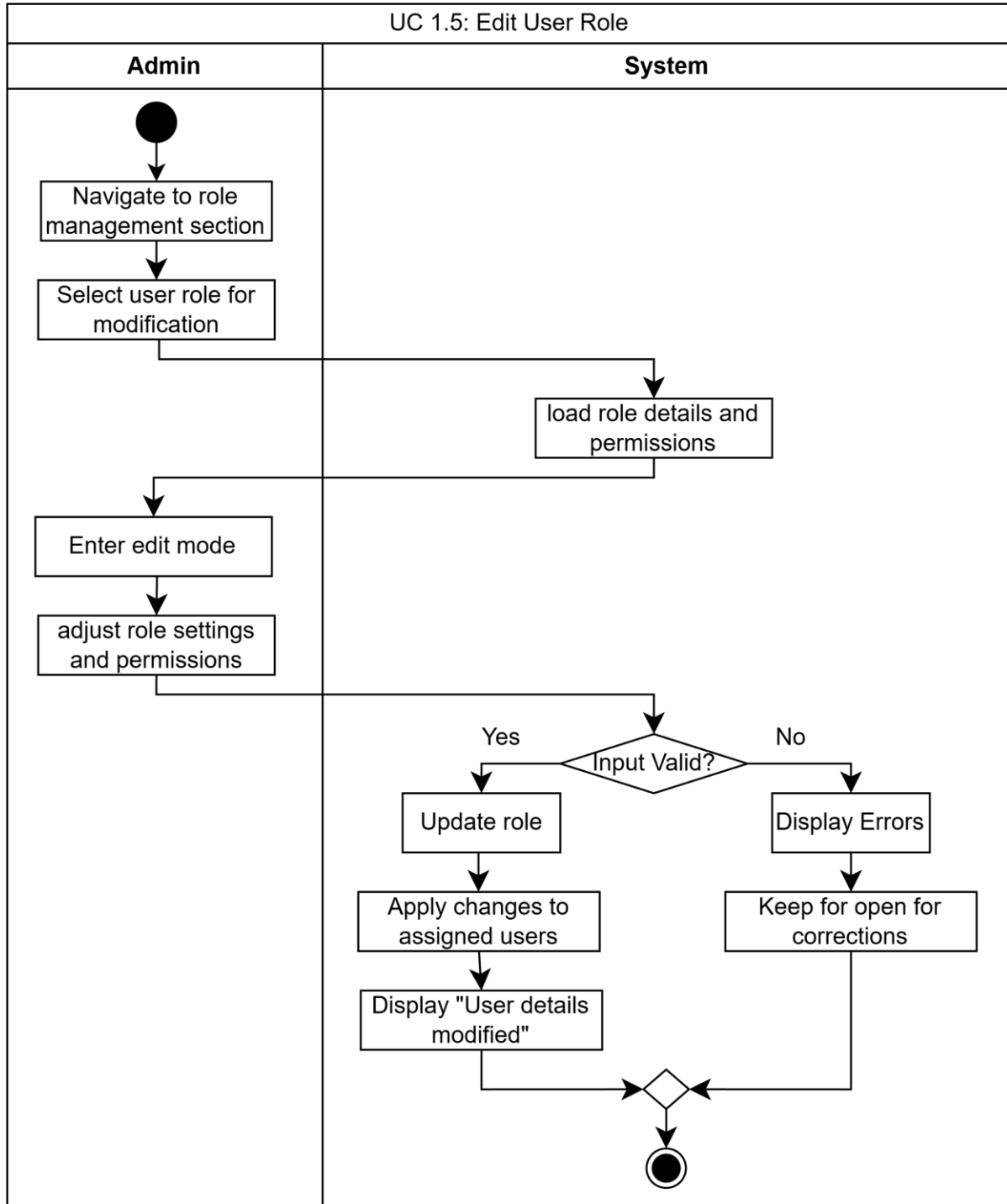
1.0 Manage Users

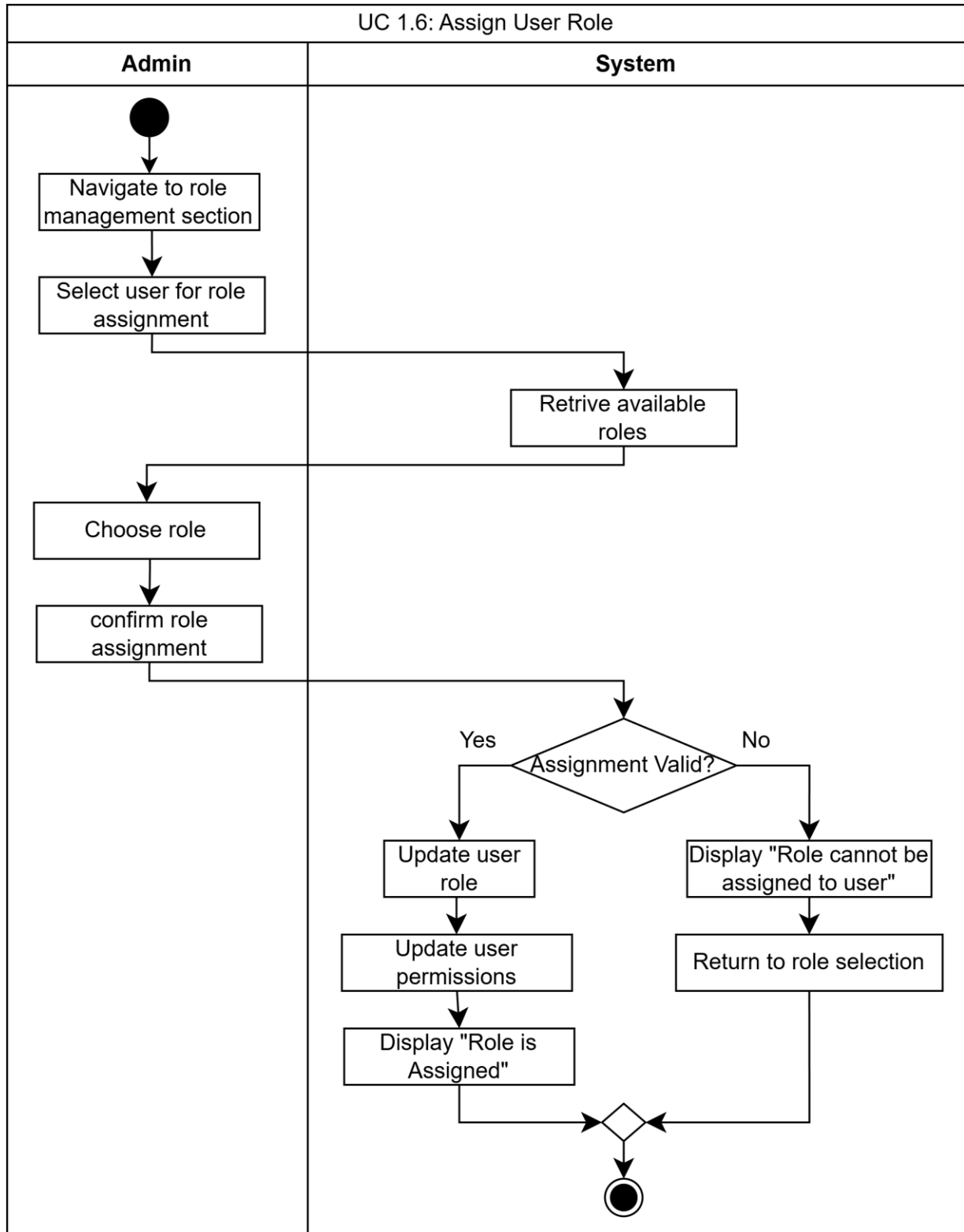


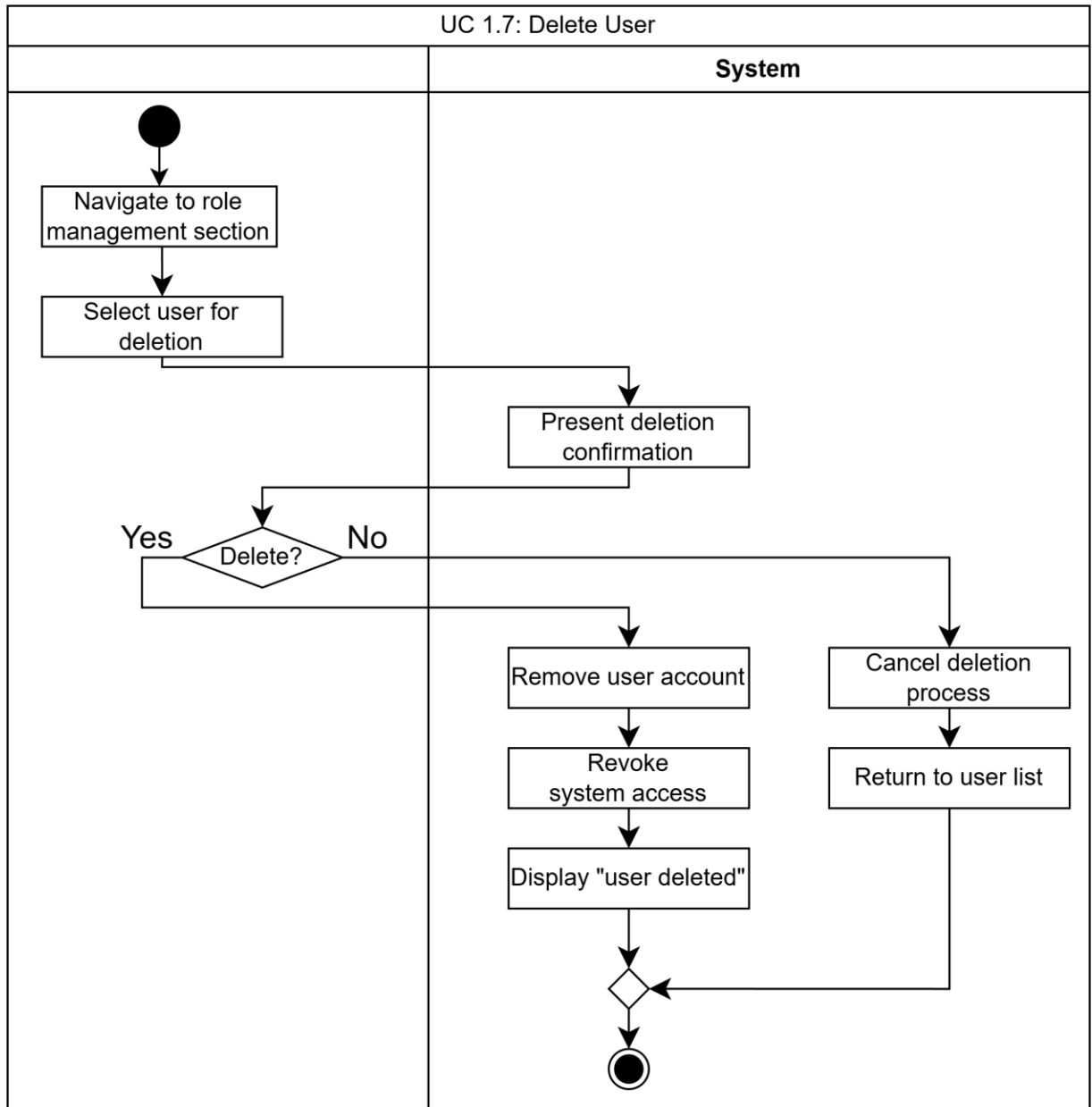


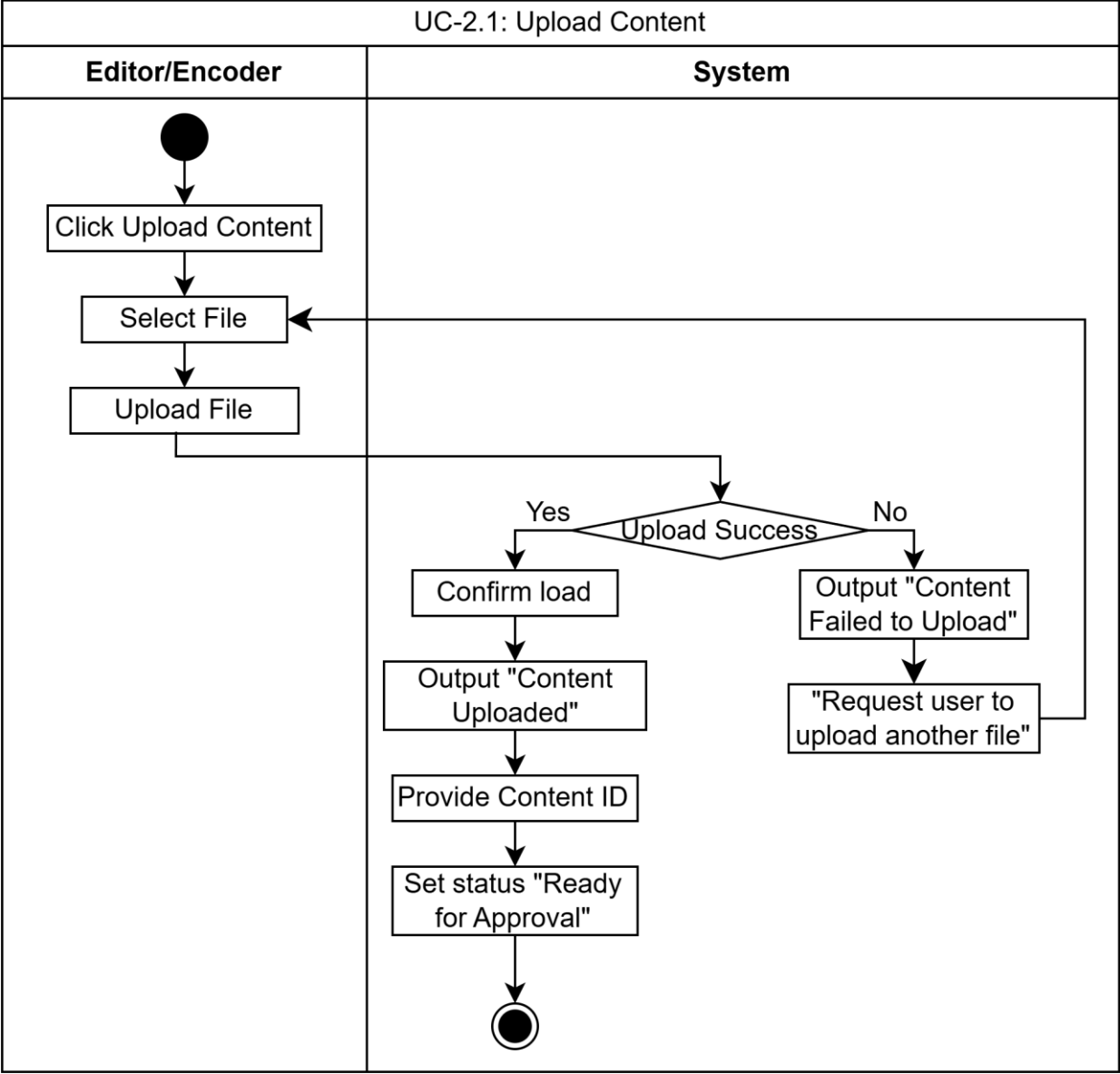


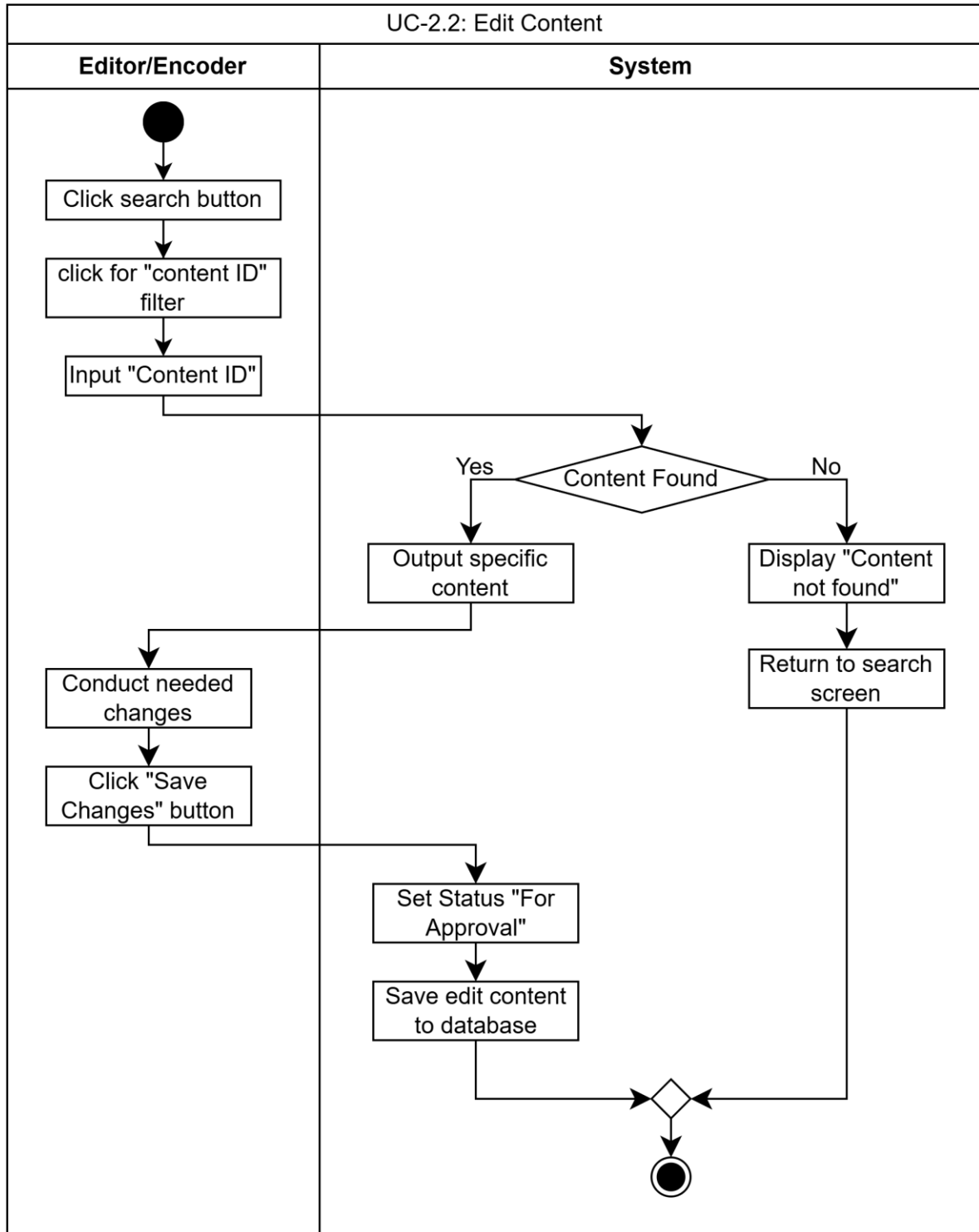


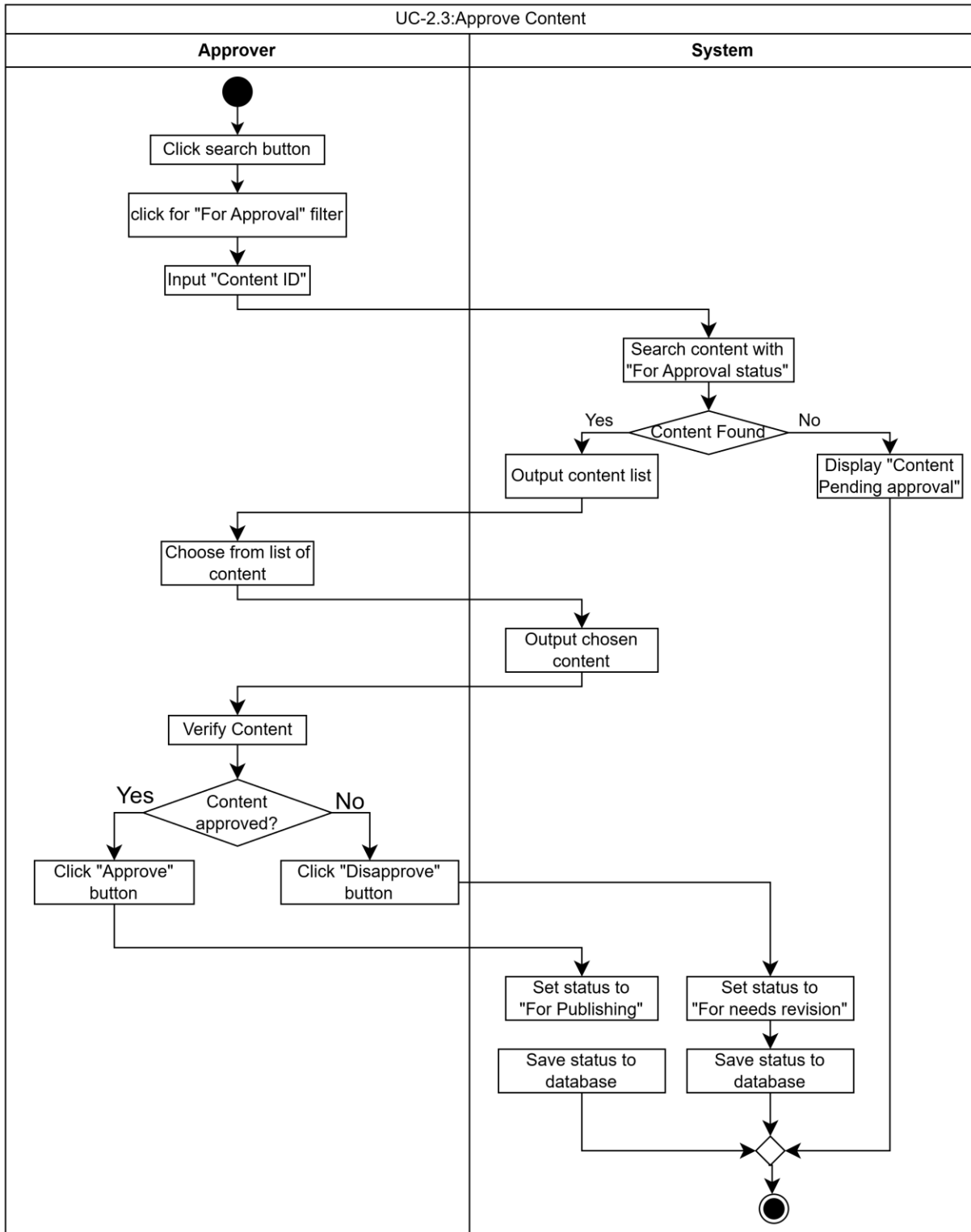


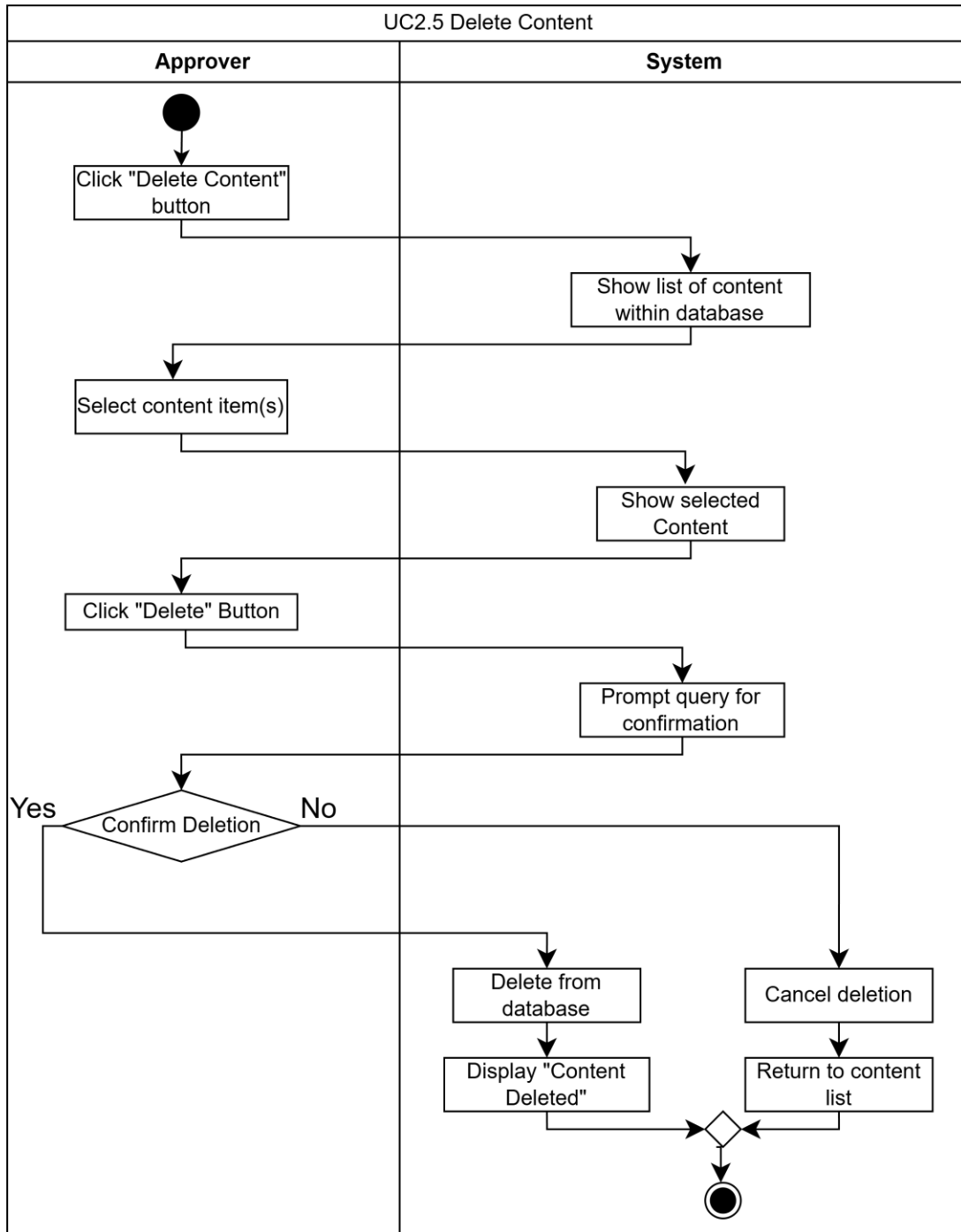




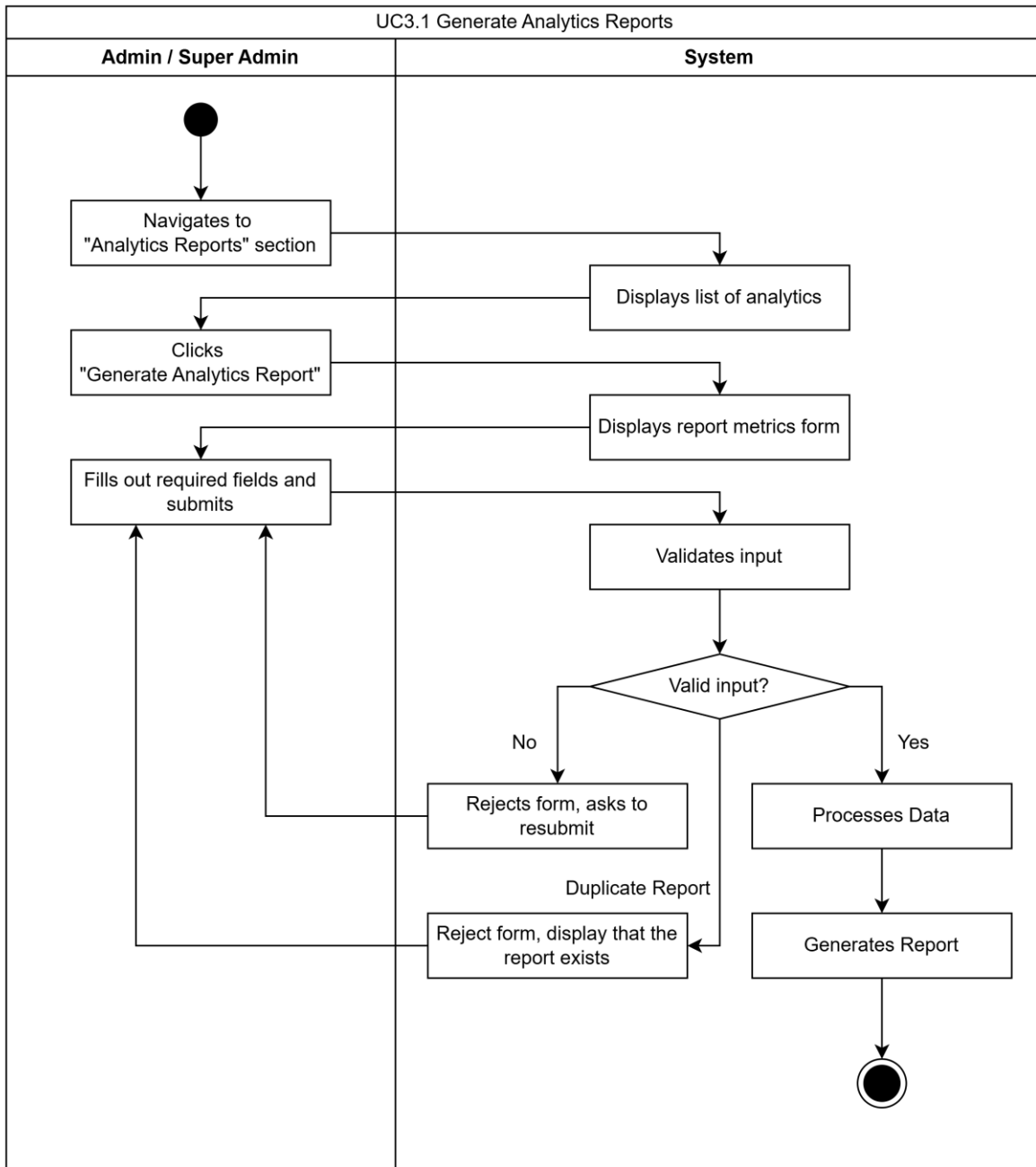


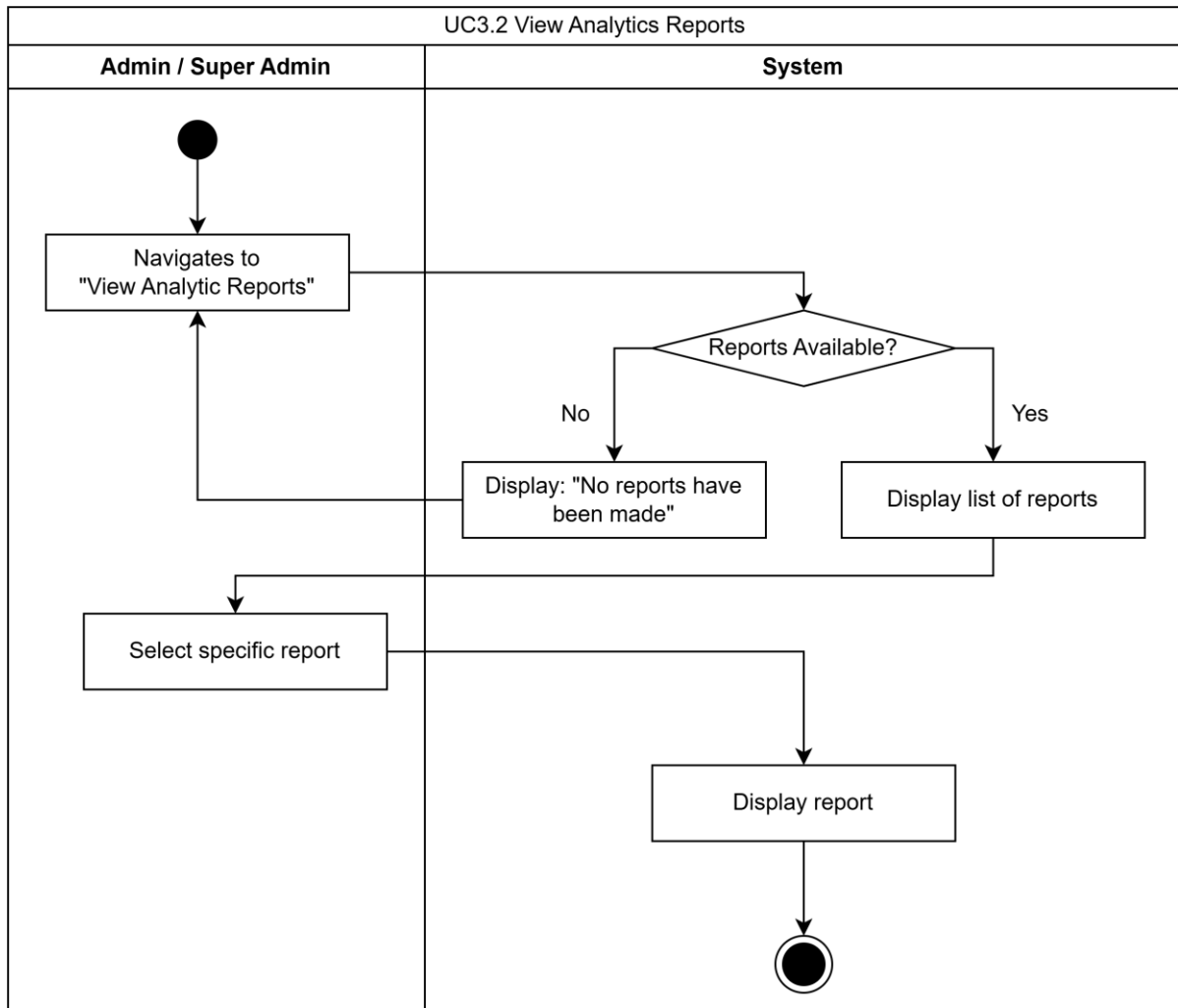


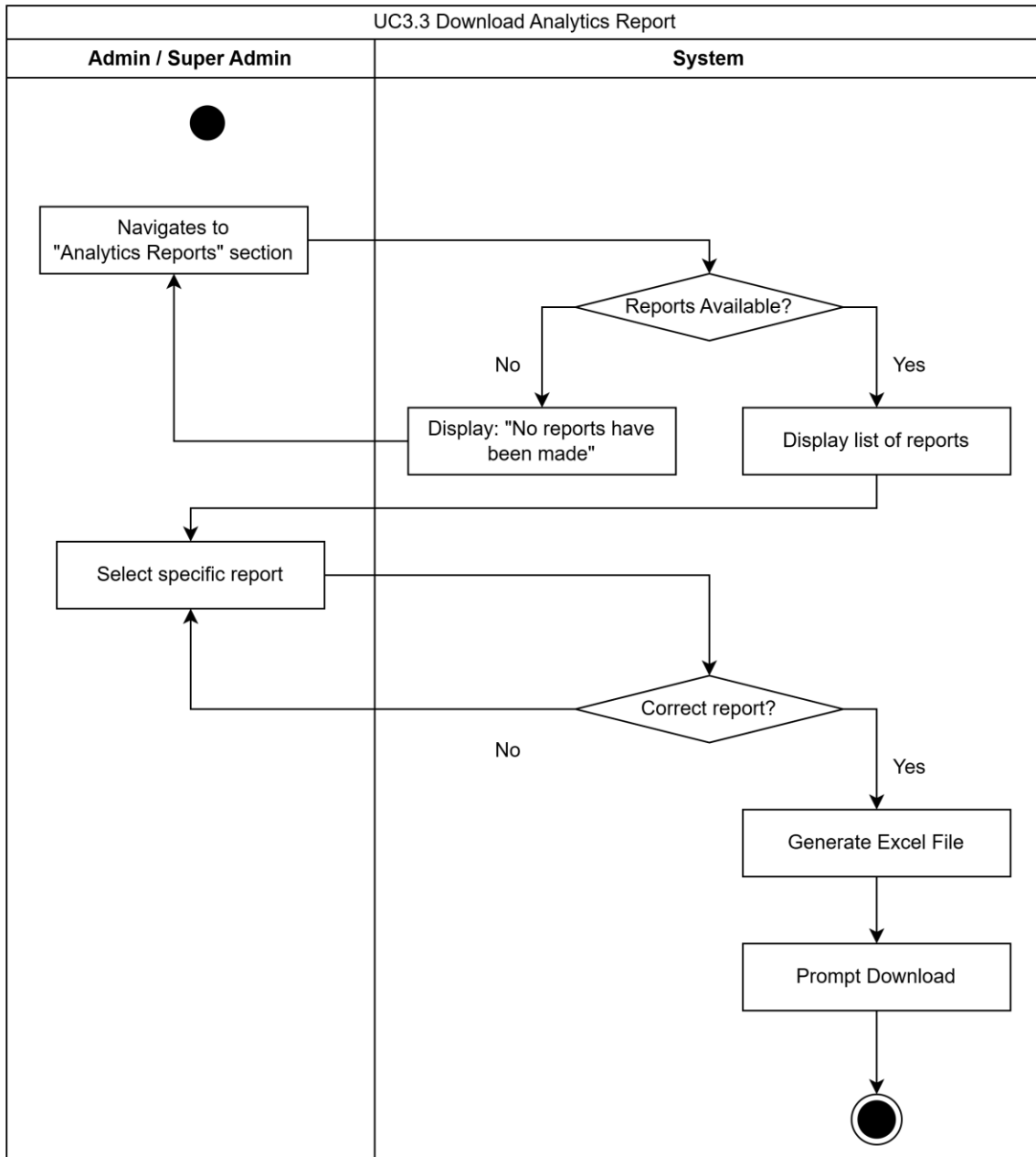




3.0 Manage Analytics Reports







Test Cases

1.0 Manage Users

Test Case ID	TC1.1.1	Test Case Description	Verify that the system allows users to log in successfully with valid credentials.			Use Case ID	UC1.1
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Authentication - Login with Valid Credentials
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	User account must already exist.		1	Username: encoder_dost			
2	Access to the CMS login page.		2	Password: Encoder123			
3			3				
4			4				
Test Scenario: Verify that the user can log in with valid credentials and is redirected to the dashboard.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to the login page	Login page loads successfully.					
2	Enter valid username and password.	System accepts entered credentials.					
3	Click the "Login" button.	User is redirected to their dashboard with the message "Login successful."					
Test Case ID	TC1.1.2	Test Case Description	Verify that the system displays an error message when invalid credentials are entered.			Use Case ID	UC1.1
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Authentication - Login with Invalid Credentials
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Access to CMS login page.		1	Username: encoder_dost			
2			2	Password: WrongPass123			
3			3				
4			4				
Test Scenario: Verify that invalid credentials display an error message and prevent login.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Go to the login page.	Login page displays correctly.					
2	Enter invalid username or password.	System detects invalid credentials.					
3	Click "Login."	System displays error message "Invalid username or password" and returns user to the login page.					
Test Case ID	TC1.1.3	Test Case Description	Verify that the system prevents login when required fields are empty.			Use Case ID	UC1.1
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Authentication - Login with Empty Fields
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Access to CMS login page.		1	Username: blank			
2			2	Password: blank			
3			3				
4			4				
Test Scenario: Verify that invalid credentials display an error message and prevent login.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Leave both username and password fields empty.	System detects missing required fields.					
2	Click "Login."	System displays a validation message: "Please enter username and password."					

Test Case ID	TC1.2.1	Test Case Description	Verify that an Admin can successfully create a new user with valid inputs.			Use Case ID	UC1.2	
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Create User - Successfully Create New User	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:			S #	Test Data			
1	Logged in as Admin or Super Admin.			1	Full Name: Juan Dela Cruz			
2	Access to "User Management" section.			2	Username: juancruz			
3				3	Email: juancruz@dost.gov.ph			
4				4	Password: Password123			
				5	Role: Encoder			
Test Scenario	Verify that Admin can create new users successfully.							
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Go to "User Management."		User Management page loads correctly.					
2	Click "Create User."		Create User form is displayed.					
3	Fill out all required fields with valid data.		System validates all input fields successfully.					
4	Click "Submit."		System saves the new user and displays confirmation message "User created successfully."					
Test Case ID	TC1.2.2	Test Case Description	Check that the system prevents creating a user with an existing username or email.			Use Case ID	UC1.2	
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Create User - Verify Error When Creating Duplicate User	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:			S #	Test Data			
1	Logged in as Admin or Super Admin.			1	Username: juancruz			
2	Access to "User Management" section.			2	Email: juancruz@dost.gov.ph			
3	A user with the same username or email already exists in the system.			3				
4				4				
Test Scenario	Verify that Admin can create new users successfully.							
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."		Form opens successfully.					
2	Click "Create User."		System validates input data.					
3	Enter details with an existing username or email.		System detects duplicate username or email.					
4	Click "Submit."		System displays error message: "Username or email already exists."					
Test Case ID	TC1.2.3	Test Case Description	Verify that the system rejects user creation if the email format is invalid.			Use Case ID	UC1.2	
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Create User - Validate Email Format on User Creation	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:			S #	Test Data			
1	Logged in as Admin or Super Admin.			1	Username: juancruz			
2	Access to "User Management" section.			2	Email: juancruz@dost.gov.ph			
3				3				
4				4				
Test Scenario	Verify that Admin can create new users successfully.							
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."		Form opens successfully.					
2	Click "Create User."		System validates input data.					
3	Enter invalid email format (missing ".com").		System identifies invalid email format.					
4	Click "Submit."		System displays validation message: "Please enter a valid email address."					

Test Case ID	TC1.2.4	Test Case Description	Ensure that all required fields must be filled before submission.			Use Case ID	UC1.2	
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User - Required Field Validation on User Creation	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:		S #	Test Data				
1	Logged in as Admin or Super Admin.		1	Full Name: blank				
2	Access to "User Management" section.		2	Username: blank				
3			3	Email: blank				
4			4					
Test Scenario: Verify that Admin can create new users successfully.								
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."		Form opens successfully.					
2	Click "Create User."		System performs validation.					
3	Leave required fields empty.		System highlights required fields in red.					
4	Click "Submit."		System displays message: "Please fill out all required fields."					
Test Case ID	TC1.2.5	Test Case Description	Verify that the system enforces password complexity requirements.			Use Case ID	UC1.2	
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User - Password Strength Validation	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:		S #	Test Data				
1	Logged in as Admin or Super Admin.		1	Password: 12345				
2	Access to "User Management" section.		2					
3			3					
4			4					
Test Scenario: Verify that Admin can create new users successfully.								
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."		Form opens successfully.					
2	Enter all required information		System validates password strength.					
3	Input a weak password (less than 8 characters).		System detects weak password.					
4	Click "Submit."		System displays message: "Password must be at least 8 characters with a mix of letters and numbers."					
Test Case ID	TC1.3.1	Test Case Description	Verify that the Admin can successfully update an existing user's information.			Use Case ID	UC1.3	
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit User - Successfully Edit Existing User Information	
QA Tester's Log								
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)				
S #	Prerequisites:		S #	Test Data				
1	Logged in as Admin or Super Admin.		1	User ID: U-1024				
2	Access to "User Management" section.		2	Old Email: juancruz@dost.gov.ph				
3	Existing user record is available.		3	New Email: juan.cruz@dost.gov.ph				
4			4	Role: Editor				
Test Scenario: Verify that Admin can create new users successfully.								
Step #	Step Details		Expected Results		Actual Results			Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."		User list loads correctly.					
2	Select an existing user from the list.		Edit User form is displayed with current user data.					
3	Click "Edit User."		System accepts the new input values.					
4	Modify email and role fields with valid values.		User details are updated in the database.					
5	Click "Save Changes."		System displays confirmation message: "User updated successfully."					

Test Case ID	TC1.3.2	Test Case Description	Verify that the system displays an appropriate error when trying to edit a user that does not exist.			Use Case ID	UC1.3
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit User - Validate Error When Editing Non-Existent User
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Logged in as Admin or Super Admin.			1	User ID: U-9999 (non-existent)		
2	Access to "User Management" section.			2			
3	User record is unavailable.			3			
4				4			
Test Scenario: Verify that Admin can create new users successfully.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Select a non-existing user.	System checks for the existence of the user.					
2	Attempt to access edit page for user ID U-9999.	System does not find user record.					
3	Click "Save Changes."	System displays error message: "User not found."					
Test Case ID	TC1.3.3	Test Case Description	Ensure the system validates input fields during edit (e.g., invalid email or empty required fields).			Use Case ID	UC1.3
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit User - Validate Field Constraints When Editing User
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Logged in as Admin or Super Admin.			1	Email: juancruz@dost		
2	Access to "User Management" section.			2			
3	Admin is editing an existing user.			3			
4				4			
Test Scenario: Verify that Admin can create new users successfully.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Edit User."	Form loads successfully.					
2	Enter an invalid email format.	System identifies invalid email format.					
3	Click "Save Changes."	Error message displayed: "Please enter a valid email address."					
4							
5							
Test Case ID	TC1.3.4	Test Case Description	Verify that a non-admin user cannot edit another user's details.			Use Case ID	UC1.3
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit User - Unauthorized Edit Attempt
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Logged in as Admin or Super Admin.			1	Username: encoder_dost		
2	Access to "User Management" section.			2	Target User: U-1024		
3	User with "Encoder" role is logged in.			3			
4				4			
Test Scenario: Verify that Admin can create new users successfully.							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in as Encoder.	Encoder dashboard loads correctly.					
2	Attempt to access "Edit User" for another user.	System restricts access to Edit User form.					
3	Modify details and click "Save."	System displays error: "Access denied. Only Admins can modify user data."					
4							
5							

Test Case ID	TC1.4.6	Test Case Description	Ensure that all user deletions are recorded in the system's audit logs.			Use Case ID	UC1.4
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete User
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	Admin or Super Admin is logged in.	1	User ID: U-1024				
2	Admin has successfully deleted a user.	2					
3		3					
4		4					
Test Scenario	Verify Audit Log Entry After Deletion						
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Delete user U-1024.	Deletion is successful.					
2	Navigate to the "Audit Logs" module.	Audit log records the deletion event with user ID, deleter's username, and timestamp.					
3	Search for entry referencing user U-1024.	Entry appears under "User Management – Delete Action."					
4							
5							
Test Case ID	TC1.5.1	Test Case Description	Verify that an Admin or Super Admin can successfully create a new user role with defined permissions.			Use Case ID	UC1.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	Admin or Super Admin is logged in.	1	Role Name: Content Reviewer				
2	Access to "Role Management" section.	2	Permissions: View Content, Approve Content				
3		3					
4		4					
Test Scenario	Successfully Create a New User Role						
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Navigate to "Role Management."	System displays the role creation form.					
2	Click "Create Role."	Admin inputs valid data.					
3	Fill in role name and select permissions.	System validates the inputs.					
4	Click "Save."	System creates the new role and displays: "Role successfully created."					
5							

Test Case ID	TC1.5.2	Test Case Description	Verify that the system prevents creation of a role with a name that already exists.			Use Case ID	UC1.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Role "Content Reviewer" already exists in the system.		1	Role Name: Content Reviewer			
2			2	Permissions: View Content, Approve Content			
3			3				
4			4				
Test Scenario	Attempt to Create a Duplicate Role						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Go to "Role Management."	System displays the role creation form.					
2	Click "Create Role."	System detects duplicate role name.					
3	Enter role name identical to an existing one.	System prevents creation.					
4	Click "Save."	Error message displayed: "Role already exists. Please use a different name."					
5							
Test Case ID	TC1.5.3	Test Case Description	Ensure that the system does not allow creation when required fields are left blank.			Use Case ID	UC1.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Admin or Super Admin is logged in.		1	Role Name: blank			
2			2	Permissions: View Content			
3			3				
4			4				
Test Scenario	Create Role with Missing Fields						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Go to "Role Management."	System displays the role creation form.					
2	Click "Create Role."	System detects empty required field.					
3	Leave the role name field empty.	Validation message displayed: "Role name is required."					
4	Click "Save."	System does not create the role.					
5							

Test Case ID	TC1.5.5	Test Case Description	Ensure that successful role creation events are recorded in the audit logs.			Use Case ID	UC1.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Create User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Admin or Super Admin has created a new role.			1	Role Name: Data Analyst		
2				2			
3				3			
4				4			
Test Scenario: Validate Role Creation Logging							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Create a new role "Data Analyst."	Role is successfully created.					
2	Navigate to "Audit Logs."	System logs creation event with role name, creator's username, and timestamp.					
3	Search for the entry corresponding to "Role Creation."	Log entry appears in the "Audit Logs" module.					
4							
5							

Test Case ID	TC1.6.2	Test Case Description	Ensure that the system prevents role updates when required fields are left blank.			Use Case ID	UC1.6
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Edit User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Admin or Super Admin is logged in.			1	Existing Role Name: Content Reviewer		
2				2	New Role Name: blank		
3				3			
4				4			
Test Scenario	Attempt to Edit Role with Empty Fields						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Role Management."	System displays list of roles.					
2	Select role "Content Reviewer."	System shows editable role details.					
3	Click "Edit."	System proceeds to the Edit form.					
4	Leave the role name empty.	System detects missing role name and validation message displayed: "Role name cannot be blank."					
5	Click "Save."	System does not update the role.					

Test Case ID	TC1.6.3	Test Case Description	Verify that the system prevents renaming a role to a name already in use.			Use Case ID	UC1.6
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Edit User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Admin or Super Admin is logged in.		1	Current Role: Content Reviewer			
2	Two existing roles: "Content Reviewer" and "Editor."		2	New Role Name: Editor			
3			3				
4			4				
Test Scenario Attempt to Rename Role to Existing Name							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Role Management."	System displays list of roles.					
2	Select role "Content Reviewer."	System shows editable role details.					
3	Click "Edit."	System proceeds to the Edit form.					
4	Enter "Editor" as the new name.	Detects duplication.					
5	Click "Save."	Displays error message: "Role name already exists." and retains previous role name.					
Test Case ID	TC1.6.4	Test Case Description	Ensure that users with non-admin roles cannot edit role configurations.			Use Case ID	UC1.6
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Edit User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Encoder is logged in.		1	Role Name: Editor			
2			2				
3			3				
4			4				
Test Scenario Unauthorized Role Edit Attempt							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in as Encoder.	System loads "Role Management" section.					
2	Navigate to "Role Management."	System checks user privileges.					
3	Attempt to edit "Editor" role.	Access denied message appears: "Access denied. Only Admins can edit roles."					
4							
5							

Test Case ID	TC1.7.4	Test Case Description	Ensure that only Admin or Super Admin can assign roles.			Use Case ID	UC1.7
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Assign User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Encoder is logged in.		1	Username: maria.editor			
2			2	Role: Reviewer			
3			3				
4			4				
Test Scenario: Unauthorized Role Assignment Attempt							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Log in as Encoder.	Successfully able to log in.					
2	Navigate to "User Management."	System loads user details.					
3	Select maria.editor.	System checks user privileges.					
4	Attempt to click "Assign Role."	Access denied message displayed: "Access denied. Only Admins can assign roles."					
5							
Test Case ID	TC1.7.5	Test Case Description	Verify that Admin can update an existing user's role.			Use Case ID	UC1.7
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Assign User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	User already has an assigned role.		1	Username: juan.editor			
2			2	Current Role: Editor			
3			3	New Role: Reviewer			
4			4				
Test Scenario: Reassign Role for Existing User							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Go to "User Management."	System loads user details.					
2	Select juan.editor.	Admin selects new role.					
3	Click "Assign Role."	System validates input.					
4	Change role from "Editor" to "Reviewer."	System updates user's role in database.					
5	Confirm update.	System displays: "Role successfully updated."					

Test Case ID	TC1.7.6	Test Case Description	Ensure that successful role assignments are recorded in the audit log.			Use Case ID	UC1.7
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Assign User Role
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Admin has successfully assigned a role to a user.		1	Username: maria.editor			
2			2	Assigned Role: Reviewer			
3			3				
4			4				
Test Scenario: Validate Audit Log After Role Assignment							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Assign the role "Reviewer" to maria.editor.	Role assignment is completed.					
2	Navigate to "Audit Logs."	System logs the event with username, role name, admin name, and timestamp.					
3	Search for the log entry related to role assignment.	Entry appears in the "Audit Logs" module under "User Management – Role Assignment."					
4							
5							

Test Case ID	TC1.8.1	Test Case Description	Verify that an Admin or Super Admin can delete an existing user account.			Use Case ID	UC1.8
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete User
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Logged in as Admin or Super Admin.		1	Username: editor_maria			
2	Access to "User Management" section.		2	User ID: U1023			
3	User record exists in the database.		3				
4			4				
Test Scenario: Successfully Delete an Existing User							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "User Management."	User details are displayed correctly.					
2	Search for editor_maria.	System confirms the delete request.					
3	Click "Delete User."	Upon confirmation, the system removes the user from the database.					
4	Confirm the deletion in the popup dialog.	Success message displayed: "User successfully deleted."					
5							

Test Case ID	TC1.8.2	Test Case Description	Ensure that the system handles deletion requests for users who are not in the database.			Use Case ID	UC1.8
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Delete User

QA Tester's Log

Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:	S #	Test Data
1	Logged in as Admin or Super Admin.	1	Username: unknown_user
2	Access to "User Management" section.	2	
3	User not found in the system.	3	
4		4	

Test Scenario: Attempt to Delete Non-Existent User

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."	System searches for user.		
2	Search for unknown_user.	System finds no matching record.		
3	Attempt to click "Delete."	Error message displayed: "User not found." and no deletion occurs.		
4				
5				

Test Case ID	TC1.8.3	Test Case Description	Verify that user deletion is canceled when the Admin clicks "Cancel" on the confirmation dialog.			Use Case ID	UC1.8
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Delete User

QA Tester's Log

Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)	
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S #	Prerequisites:	S #	Test Data
1	Logged in as Admin or Super Admin.	1	Username: juan_reviewer
2	Access to "User Management" section.	2	
3		3	
4		4	

Test Scenario: Cancel Deletion During Confirmation

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Navigate to "User Management."	User list displays correctly.		
2	Select juan_reviewer.	System prompts confirmation.		
3	Click "Delete User."	Admin cancels the operation.		
4	In confirmation popup, click "Cancel."	System aborts the delete process and user record remains unchanged.		
5				

Test Case ID	TC1.8.6	Test Case Description	Ensure that all user deletions are recorded in the system's audit logs.			Use Case ID	UC1.8
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Delete User
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	Admin or Super Admin is logged in.	1	Deleted User: editor_maria				
2	Admin has successfully deleted a user.	2	Admin Username: admin_dost				
3		3					
4		4					
Test Scenario: Verify Audit Log After Successful Deletion							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Delete user editor_maria.	System logs deletion event.					
2	Navigate to the "Audit Logs" module.	Log entry includes Admin username, deleted user ID, date/time, and action details.					
3	Search for "User Deletion" event.	Entry visible under "User Management – Deletion Logs."					
4							
5							
Test Case ID	TC2.1.1	Test Case Description	Verify that an Encoder or Editor can upload new content successfully.			Use Case ID	UC2.1
Created By	Frances Sophia Lepspot	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Upload Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	User is logged in as Encoder or Editor.	1	Content Title: Science in Everyday Life				
2		2	File: science_video.mp4				
3		3	Category: Science				
4		4					
Test Scenario: Successfully Upload Content							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Navigate to "Content Management."	Upload form displays correctly.					
2	Click "Upload Content."	File successfully attached.					
3	Fill in the content details.	System validates all fields.					
4	Attach science_video.mp4.	Content is uploaded and saved to the database.					
5	Click "Submit."	System displays: "Content uploaded successfully."					

Test Case ID	TC2.1.2	Test Case Description	Ensure system prevents uploading duplicate content with the same title or file.			Use Case ID	UC2.1
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Upload Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Content titled Science in Everyday Life already exists.		1	Content Title: Science in Everyday Life			
2			2	File: science_video.mp4			
3			3				
4			4				
Test Scenario: Attempt to Upload Duplicate Content							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Content Management."	Upload form displays correctly.					
2	Click "Upload Content."	File successfully attached.					
3	Enter title Science in Everyday Life.	System detects duplicate title/file.					
4	Upload science_video.mp4.	Upload fails.					
5	Click "Submit."	Error message displayed: "Content already exists. Please use a different title or file."					
Test Case ID	TC2.1.3	Test Case Description	Verify that the system displays an error when mandatory fields are left blank.			Use Case ID	UC2.1
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Upload Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	User is logged in.		1	Content Title: (blank)			
2			2	File: science_video.mp4			
3			3				
4			4				
Test Scenario: Upload with Missing Required Fields							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Content Management."	Upload form displays correctly.					
2	Click "Upload Content."	File successfully attached.					
3	Leave title field blank.	Error message displayed: "Title is required."					
4	Upload file.	Upload fails.					
5	Click "Submit."	Upload is not processed.					

Test Case ID	TC2.2.1	Test Case Description	Verify that Editor or Approver can edit content metadata.			Use Case ID	UC2.2
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Content exists in "For Editing" status.			1	Old Title: Science in Everyday Life		
2				2	New Title: Science in Motion		
3				3			
4				4			
Test Scenario	Successfully Edit Content Details						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Content Management."	List of contents displays correctly.					
2	Select Science in Everyday Life.	Form loads successfully.					
3	Click "Edit."	Edit form loads successfully.					
4	Change title to Science in Motion.	System validates new title.					
5	Click "Save."	Message displayed: "Content successfully updated."					
Test Case ID	TC2.2.2	Test Case Description	Verify that unauthorized users cannot edit content.			Use Case ID	UC2.2
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Edit Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	User logged in as Encoder.			1	Content Title: Physics Fun Facts		
2				2			
3				3			
4				4			
Test Scenario	Attempt to Edit Without Permissions						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in as Encoder.	Content Management form displays correctly.					
2	Go to "Content Management."	List of contents displays correctly.					
3	Select Physics Fun Facts.	List of contents displays correctly.					
4	Attempt to click "Edit."	Edit form loads successfully.					

Test Case ID	TC2.2.3	Test Case Description	Ensure system validates edited content before saving.			Use Case ID	UC2.2
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Edit Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	User logged in as Editor.		1	Content Title: Science in Motion			
2			2	Description: (blank)			
3			3				
4			4				
Test Scenario Attempt to Save Invalid Edits							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Go to "Content Management."	List of contents displays correctly.					
2	Select Science in Motion.	Edit form loads successfully.					
3	Click "Edit."	System validates fields.					
4	Clear the Description field.	Validation fails for empty description.					
5	Click "Save."	Error message displayed: "Description cannot be empty."					
Test Case ID	TC2.3.1	Test Case Description	Verify that Approver can approve content for publishing.			Use Case ID	UC2.3
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Approve Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Content status is "For Approval."		1	Content Title: Science in Motion			
2	User logged in as Approver.		2				
3			3				
4			4				
Test Scenario Successfully Approve Content							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Go to "Content Management."	List of contents displays correctly.					
2	Select Science in Motion.	System validates user role.					
3	Click "Approve."	Status changes to "Approved."					
4	Confirm approval.	Message displayed: "Content approved successfully."					

Test Case ID	TC2.3.2	Test Case Description	Verify that Approver can reject content and return it for editing.			Use Case ID	UC2.3
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Approve Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	Content status is "For Approval."			1	Content Title: Science in Motion		
2				2	Remarks: Add more visuals.		
3				3			
4				4			
Test Scenario: Reject Content for Revision							
Step #	Step Details		Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
1	Go to "Content Management."		List of contents displays correctly.				
2	Select Science in Motion.		System validates user role.				
3	Click "Reject."		System records remarks.				
4	Enter remarks "Add more visuals."		Content status changes to "For Editing."				
5	Confirm action.		Notification sent to Encoder and message displayed: "Content returned for revision."				
Test Case ID	TC2.3.3	Test Case Description	Ensure remarks are required when rejecting content.			Use Case ID	UC2.3
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Approve Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	User logged in as Approver.			1	Content Title: Science in Motion		
2				2	Remarks: (blank)		
3				3			
4				4			
Test Scenario: Attempt to Approve Without Remarks							
Step #	Step Details		Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
1	Go to "Content Management."		List of contents displays correctly.				
2	Select Science in Motion.		System validates user role.				
3	Click "Reject."		System validates remarks field.				
4	Leave remarks blank.		Validation fails.				
5	Confirm rejection.		Error message displayed: "Remarks required to reject content."				

Test Case ID	TC2.4.1	Test Case Description	Verify that Admin or Super Admin can successfully delete a content record.			Use Case ID	UC2.4
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	User logged in as Admin or Super Admin			1	Content Title: Science in Motion		
2	Target content exists in the system.			2			
3				3			
4				4			
Test Scenario	Successfully Delete Content						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Content Management."	Confirm deletion in the pop-up dialog.					
2	Locate Science in Motion in the content list.	User confirms deletion.					
3	Click "Delete."	System deletes the record from the database.					
4	onfirm deletion in the pop-up dialog.	Message displayed: "Content successfully deleted."					
5							
Test Case ID	TC2.4.2	Test Case Description	Ensure that if user cancels the delete confirmation, content remains intact.			Use Case ID	UC2.4
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:			S #	Test Data		
1	User logged in as Admin or Super Admin			1	Content Title: Science in Motion		
2	Content available in the list.			2			
3				3			
4				4			
Test Scenario	Cancel Content Deletion						
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Navigate to "Content Management."	Confirm deletion in the pop-up dialog.					
2	Locate Science in Motion in the content list.	User cancels deletion.					
3	Click "Delete."	Content remains in the list.					
4	Click "Cancel" on confirmation prompt.	Message displayed: "Deletion canceled."					
5							

Test Case ID	TC2.4.3	Test Case Description	Verify that only Admins or Super Admins can delete content.			Use Case ID	UC2.4
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Encoder logged in.		1	Content Title: Science in Motion			
2	Content available in the list.		2				
3			3				
4			4				
Test Scenario Attempt to Delete Without Proper Role							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in as Encoder.	System checks user permissions.					
2	Go to "Content Management."	Access denied.					
3	Try to click "Delete" on any content item.	Error displayed: "You are not authorized to delete content."					
4							
5							
Test Case ID	TC2.4.4	Test Case Description	Ensure that deleted content is permanently removed from the system database and cannot be retrieved.			Use Case ID	UC2.4
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Delete Content
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Admin has deleted content titled Science in Motion.		1	Content Title: Science in Motion			
2			2				
3			3				
4			4				
Test Scenario Validate Deleted Content Removal from Database							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Delete content Science in Motion.	Content no longer appears in the list.					
2	Refresh the content list.	Database confirms record deletion.					
3	Search for Science in Motion.	No retrieval possible through search.					
4							
5							

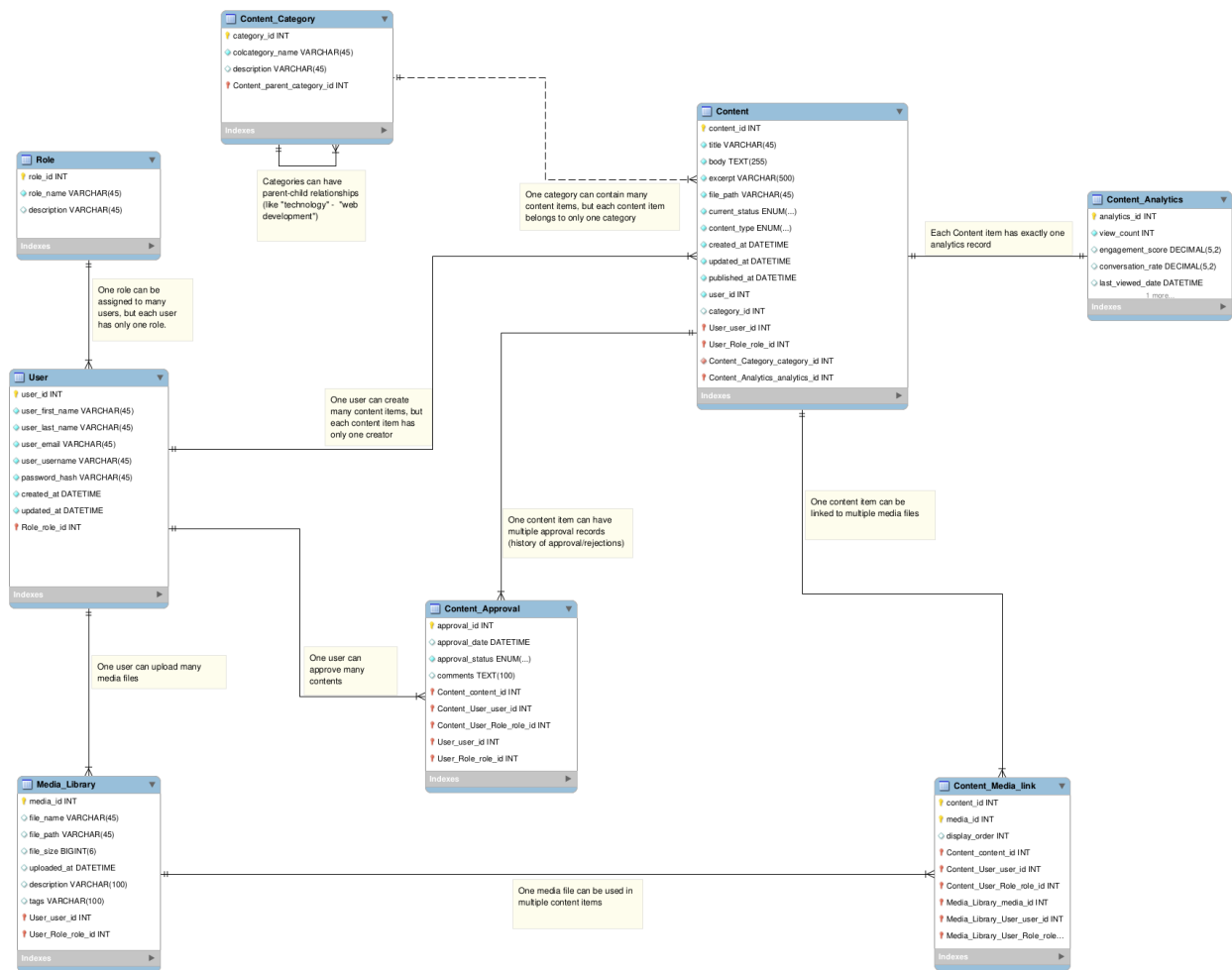
Test Case ID	TC2.5.1	Test Case Description	Verify that authorized users can view version history of a content item.			Use Case ID	UC2.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	View Content History
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	User logged in.		1	Content Title: Science in Motion			
2	Content has multiple edit versions.		2				
3			3				
4			4				
Test Scenario Successfully View Content History							
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended
1	Go to "Content Management."		System displays version history page.				
2	Select Science in Motion.		User sees timestamps, editors, and remarks of past versions.				
3	Click "View History"		Message displayed: "Displaying version history."				
4							
5							
Test Case ID	TC2.5.2	Test Case Description	Ensure that only authorized roles can view content history.			Use Case ID	UC2.5
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	View Content History
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Encoder logged in.		1	Content Title: Science in Motion			
2			2				
3			3				
4			4				
Test Scenario Attempt to View History Without Permission							
Step #	Step Details		Expected Results		Actual Results		Pass / Fail / Not executed / Suspended
1	Log in as Encoder.		System displays Content Management page.				
2	Navigate to "Content Management."		System displays lists of contents.				
3	Select Science in Motion.		System checks role privileges.				
4	Attempt to click "View History."		Access denied and error displayed: "You do not have permission to view content history."				
5							

Test Case ID	TC2.5.3	Test Case Description	Ensure that content versions are displayed in chronological order (latest first).			Use Case ID	UC2.5
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	View Content History
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #Prerequisites:				S #Test Data			
1Multiple versions of Science in Motion exist.				1Version Records (v1 – Initial Upload, v2 – Edited Description, v3 – Final Approved)			
2				2			
3				3			
4				4			
Test ScenarioVerify Correct Version Ordering							
Step #		Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
1		Navigate to “Content Management.”	System displays Content Management page.				
2		Select Science in Motion.	System retrieves all versions.				
3		Click “View History.”	Displays list in correct order (v3 → v2 → v1) and shows version details with timestamps and editor names.				
4							
Test Case ID	TC2.5.4	Test Case Description	Ensure that users can search and filter content history records.			Use Case ID	UC2.5
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	View Content History
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #Prerequisites:				S #Test Data			
1Content has at least three previous versions.				1Filter: Date: October 2025			
2				2Keyword: Approved			
3				3			
4				4			
Test ScenarioVerify History Search and Filter Function							
Step #		Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended	
1		Go to “View History” page.	System filters results correctly.				
2		Use search bar to filter by keyword “Approved.”	Only relevant history entries appear.				
3		Apply date filter for October 2025.	Message displayed: “Filtered results displayed.”				
4							

Test Case ID	TC3.1.1	Test Case Description	Verify that an authorized administrator can successfully access the analytics dashboard.			Use Case ID	UC3.0
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Manage Analytics Report
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Super admin account must exist and be logged in.		1	Username: admin01			
2			2	Password: Admin@123			
3			3				
4			4				
Test Scenario Access Analytics Dashboard (Authorized User)							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in using admin credentials.	System logs in successfully.					
2	Navigate to "Analytics" in the main menu.	Dashboard loads with summary charts and key metrics.					
3	Observe the displayed dashboard.	Data visualization components are correctly displayed.					
4							
Test Case ID	TC3.1.2	Test Case Description	Verify that users without analytics permissions are denied access.			Use Case ID	UC3.0
Created By	Frances Sophia Lepso	Reviewed By	Eryne Anthony Chan	Version	1.0	Test Case Name	Manage Analytics Report
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:		S #	Test Data			
1	Regular Encoder account must exist and be logged in.		1	Username: encoder01			
2			2	Password: Encode@123			
3			3				
4			4				
Test Scenario Attempt to Access Analytics Without Permission							
Step #	Step Details	Expected Results	Actual Results		Pass / Fail / Not executed / Suspended		
1	Log in using encoder credentials.	System denies access.					
2	Attempt to navigate to "Analytics."	Error message "Access Denied – Insufficient Permissions" appears.					
3							
4							

Test Case ID	TC3.1.3	Test Case Description	Verify that an authorized admin can generate a report for a specific date range.			Use Case ID	UC3.0
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Manage Analytics Report
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	Admin is logged in and dashboard is accessible.	1	Date Range: 2025-10-01 to 2025-10-31				
2		2	Report Type: "User Engagement"				
3		3					
4		4					
Test Scenario: Generate Report Successfully							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Go to "Analytics" dashboard.	User Engagement page displays correctly.					
2	Select "User Engagement" report.	Input data range displays correctly.					
3	Input date range.	System accepts input and processing indicator appears.					
4	Click "Generate Report."	Report displays relevant data and graphs for the given date range.					
Test Case ID	TC3.1.4	Test Case Description	Verify system response when an invalid date range or filter is provided.			Use Case ID	UC3.0
Created By	Frances Sophia Lepso	Reviewed By	Ervyne Anthony Chan	Version	1.0	Test Case Name	Manage Analytics Report
QA Tester's Log							
Tester's Name		Date Tested		Test Case (Pass/Fail/Not Executed)			
S #	Prerequisites:	S #	Test Data				
1	Super Admin is logged in.	1	Date Range: 2025-12-31 to 2025-01-01 (end date before start date)				
2	Has access to the Analytics page.	2					
3		3					
4		4					
Test Scenario: Handle Invalid Filter or Date Range Input							
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended			
1	Go to "Analytics" dashboard.	List of reports displays correctly.					
2	Select any report type.	Input data range displays correctly.					
3	Input invalid date range.	System detects invalid range.					
4	Click "Generate Report."	Error message appears: "Invalid date range. Please correct and try again."					

Entity Relationship Diagram



Technology Stack

1.1 Frontend Framework

React

The team chose React as it's written in JavaScript and it's a component-based architecture which allows us to create modular and reusable UI elements. As the website has dynamic interfaces, the virtual DOM of React ensures quick rendering and responsiveness. The frontend also communicates to the backend via RESTful APIs.

1.2 Backend Framework

Django + Wagtail CMS

The team chose Django, which is enhanced by Wagtail CMS, to provide good content management experience. Django is written in python which is what we are most familiar with and because Django being a high-level python framework which is known for its scalability. This combination allows for structured content management, user authentication, and seamless integration with custom models and APIs. The backend also includes Django REST Framework (DRF) which exposes data via RESTful endpoints. For authentication, Simple JWT is used.

1.3 Frontend / Backend Communication

RESTful API

The frontend and backend communicate through RESTful API calls. The CORS headers are configured on the backend to allow cross-origin requests to be processed securely.

GitHub APC-SoCIT Project Repository: [\[link\]](#)

OpenProject Team Site: [\[link\]](#)

Cloud Hosted Site (Django Admin Page): [\[link\]](#)

Cloud Hosted Site (AGHAMazingQuestCMS Dashboard): [\[link\]](#)